

Technical Design Document

LinkNet Corporation, Enterprise, Fiber, Media

Item	Value
Document title	Technical Design Document - LinkNet Corporation
Project name	LinkNet Corporation, Enterprise, Fiber, Media
Version	1.0
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Prepared by	Media Indo Trend

1. Document Information

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Document title	Technical Design Document - LinkNet Corporation
Project name	LinkNet Corporation
Version	1.0
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2. Project Overview

Aplikasi ini adalah platform corporate web dan CMS LinkNet Corp berbasis beberapa aplikasi dalam satu workspace. Implementasi aktual terdiri dari backend REST API, admin CMS dashboard, public website, dan service file manager terpisah. Dokumen ini disusun dari source code aktual, route definitions, Prisma schema, konfigurasi, middleware, Dockerfile, dan workflow CI yang ditemukan.

Aspect	Description
Tujuan aplikasi	Menyediakan CMS dan API untuk pengelolaan corporate website, content modules, dynamic pages, user/RBAC, file management, form submissions, analytics dashboard, dan public web consumption.
Ruang lingkup sistem	Backend API, Admin CMS frontend, Public web frontend, standalone file manager service, database PostgreSQL melalui Prisma, object/blob storage integration, analytics and external proxy integrations.
High level description	Admin user mengelola content melalui frontend Next.js yang memanggil Express API. Public web Next.js membaca content CMS dan melakukan submit form ke API. Backend menyimpan data di PostgreSQL dan file metadata/storage melalui local/S3/Azure-related services.
Business process summary	Login CMS -> role/permission check -> manage content/config/files/forms -> publish/serve content to public web -> collect submissions/visits/consents -> dashboard/log review.

3. Technology Stack

Component	Package name	Version	Dependencies	Dev dependencies	Key packages found
Backend API	linknetcorp-backend	1.0.0	51	27	express@^4.21.3, @prisma/client@^5.7.1, prisma@^5.22.0 (dev), typescript@^5.3.3 (dev), jsonwebtoken@^9.0.2, bcryptjs@^2.4.3, helmet@^7.1.0, cors@^2.8.5, express-rate-limit@^7.1.5, multer@^2.0.3, @aws-sdk/client-s3@^3.1001.0, @azure/storage-blob@^12.29.1, @azure/keyvault-secrets@^4.10.0, @google-analytics/data@^5.2.1, bull@^4.16.5, ioredis@^5.3.2
Admin Frontend	free-nextjs-admin-dashboard	2.2.2	45	11	next@16.2.4, react@^19.2.0, react-dom@^19.2.0, tailwindcss@^4.1.17 (dev), axios@^1.15.0, lucide-react@^0.562.0, @svgr/webpack@^8.1.0 (dev)
Public Web	linknet_web	0.1.0	12	12	next@16.2.4, react@19.2.3, react-dom@19.2.3, next-intl@^4.9.1, gsap@^3.14.2, highcharts@^12.5.0, swiper@^12.1.3, yahoo-finance2@^3.14.0
Standalone File Manager	filemanager	1.0.0	10	11	express@^4.19.2, @aws-sdk/client-s3@^3.600.0, @aws-sdk/s3-request-presigner@^3.600.0, helmet@^7.1.0, cors@^2.8.5, express-rate-limit@^7.3.1,

					multer@^2.0.0, typescript@^5.4.5 (dev)
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Category	Implementation Found
Backend framework	Express.js with TypeScript.
Frontend framework	Next.js with React; admin app under frontend and public website under web.
Database	PostgreSQL via Prisma datasource provider postgresql.
Authentication	JWT access token + refresh token rotation; HttpOnly cookies and JSON token response; bcrypt password hashing; optional TOTP MFA.
Infrastructure	Dockerfiles for backend, frontend, web, and filemanager; Kubernetes manifests for backend; GitHub Actions CI and Trivy scan.
Third-party services	GA4 Data API, AWS S3, Azure Blob/Key Vault, Yahoo Finance, Linknet Enterprise/Media upstream proxies, optional Redis/Bull queue, email via nodemailer service code.
Cloud/storage integration	S3 service implemented for upload/list/delete/presigned; Azure Blob service implemented; local upload static path supported; storage abstraction S3 provider marked incomplete in code.

4. System Architecture

The current architecture is a multi-application workspace. The backend exposes REST APIs under `/api/v1` and health/root endpoints outside the prefix. The admin frontend is a protected CMS dashboard. The public web app renders CMS-driven pages, news, careers, events, dynamic forms, and market data. A separate filemanager service exposes an S3-backed API under `/api` with optional API key authentication.

Layer	Actual implementation
Client/Admin	frontend/src/app Next.js app, AuthContext, service layer with credentials: include, Bearer token support, CSRF header support.
Client/Public	web/app Next.js app, next-intl middleware, cmsApi/formsApi/eventsApi, visitor tracker, local API routes for Yahoo Finance and form proxy.
API	backend/src/server.ts Express app with security, CORS, CSRF, rate limit, activity logging, route modules, global error handler.
Persistence	backend/prisma/schema.prisma models and migrations; Prisma client singleton under backend/src/config/database.ts.
Storage	Uploads served from UPLOAD_DIR for local; S3 and Azure service implementations found; file metadata persisted in File/Folder tables.
Async/Queue	Bull + Redis for activity/form dispatch when REDIS_ENABLED=true; fallback direct logging for activity logger.

4.1 Frontend-Backend Communication

- Admin frontend uses `NEXT\_PUBLIC\_API\_URL`, API prefix `/api/v1`, `credentials: include`, Authorization bearer token from localStorage/cookie, and `X-CSRF-Token` from cookie for unsafe methods.
- Public web uses `NEXT\_PUBLIC\_API\_URL` or server-side `API\_INTERNAL\_URL` for CMS content, form modules, events, menu, settings, labels, and news.
- Next.js web API routes proxy Yahoo Finance data and enterprise form submission.

4.2 Authentication Flow

- Login receives email/password, applies business-hour middleware and login rate limiter, checks lockout/status/password expiry, verifies bcrypt hash, optionally requires MFA, then issues access and refresh JWTs.
- Refresh token is persisted as SHA-256 hash in `refresh\_tokens` and rotated on refresh.
- Access token may be sent via Authorization header or `auth\_token` HttpOnly cookie. Cookie-authenticated unsafe API methods require matching `X-CSRF-Token` and `csrf\_token` cookie.
- RBAC middleware checks role/permission before protected CMS operations.

4.3 Request Flow

`Client request -> HTTPS redirect middleware -> Helmet/custom security headers -> request id -> HTTP logger -> CORS -> body parser/cookie parser -> CSRF middleware on /api -> compression -> health/static handling -> no-store + rate limiter + activity logger -> route middleware/controller -> Prisma/service/storage/external integration -> global error handler`.

4.4 Deployment Overview

Docker multi-stage builds are present for all four deployable components. Backend Kubernetes manifests include Deployment, Service, HPA, probes, securityContext, ConfigMap/Secret references, ingress TLS, and Azure Key Vault CSI examples. No complete docker-compose file was found in the repository root.

5. Folder Structure Explanation

Path	Purpose
backend/	Express/TypeScript API, Prisma schema/migrations, controllers, routes, middleware, services, storage integrations, Docker/Kubernetes files.
backend/src/routes/	Mounted REST route modules used by server.ts.
backend/src/controllers/	Request handlers for authentication, CMS modules, public content, file management, dashboards, and proxies.
backend/src/middleware/	Authentication, RBAC, CSRF, rate limiting, logging, upload, security headers, error handling, environment validation.
backend/src/services/	Business/integration services including activity logging, analytics, storage/cloud, email, account lifecycle, token cleanup.
backend/prisma/	Prisma schema, migrations, and manual SQL migrations.
frontend/	Next.js admin/CMS dashboard with protected route middleware, auth context, services, and admin pages.
web/	Next.js public website with localized routing, CMS API consumers, dynamic pages, public API proxy routes, and UI components.
filemanager/	Separate Express service for S3 file operations protected by optional API key.
shared/	Shared presentation/static data found in current codebase.
docs/	Project documentation artefacts including this generated TDD.
.github/workflows/	CI build workflow and Trivy vulnerability/secret scanning workflow.

6. Module Design

Module	Description	Controllers/Services/Routes	Related Models	Data Flow	Security Consideration
Authentication & Account Security	Login/register/refresh/logout/password reset/MFA/profile session validation.	auth.controller.ts, mfa.controller.ts, auth.routes.ts, auth.middleware.ts, jwt.util.ts	User, RefreshToken, PasswordResetToken, PasswordHistory	Client -> /auth -> validators/rate limiter -> controller -> Prisma -> JWT/cookies	Bcrypt, JWT secrets validation, refresh rotation, account lockout, MFA optional, CSRF for cookie auth.
User, Role & Permission	CMS user CRUD and RBAC administration.	user.controller.ts, role.controller.ts, user.routes.ts, role.routes.ts, rbac.middleware.ts	User, Role, Permission, UserRole, RolePermission	Admin -> protected endpoint -> auth -> permission middleware -> Prisma	Permission middleware on routes; validators check UUID, status, roles.
Settings & Menu	Public and CMS configuration plus navigation hierarchy.	settings.routes.ts, menu.routes.ts, SettingsController, MenuController	Setting, Menu	CMS edits config/menu -> public web consumes /settings/public and /menu	Protected CMS routes; public read-only endpoints; potential frontend/backend endpoint mismatch noted.
Pages, Components & Labels	Page builder, component data, visibility, and label data bank.	component.routes.ts, page.routes.ts, componentVisibility.routes.ts, labelDataBank.routes.ts	Page, PageComponent, ComponentVisibility, LabelGroup, LabelNode	Admin creates page/component -> public /pages reads published content	CMS routes require auth/permissions; component route order is intentionally before page routes to avoid conflict.
Content Modules	News, events, careers, reports, announcements, awards, and management content.	news/event/career/report/announcement/award/management routes/controllers	news*, events*, Career, reports, announcements, Award, Management	CMS CRUD/order -> public list/detail/filter endpoints -> web pages	Protected CMS routes; public write endpoints for registrations/views/submissions should remain rate limited.
Forms & Contact	Dynamic forms, public submissions, file attachments, contact submissions, CMS review/export.	formModule.routes.ts, formModuleCms.routes.ts, contact.routes.ts, cms/contactus.routes.ts	FormModule and related form tables, ContactSubmission, ContactUs	Public submit -> validation/service -> DB/files/dispatch logs -> CMS review/export	Zod validation; upload middleware; CMS auth. Route order issue found for contact export.
File Management & Upload	Unified upload, file manager, storage listing/delete, presigned URLs.	upload.routes.ts, file.routes.ts, filemanager.routes.ts, filemanagerV2.routes.ts, storage/s3/azure services	File, Folder	Admin upload -> multer validation/scanning -> storage provider/service -> DB metadata	MIME/extension/size validation, optional ClamAV scanner, RBAC; storage abstraction S3 provider incomplete.
Dashboard, Analytics &	Visitor tracking, dashboard statistics,	dashboard.routes.ts, analytics.routes.ts, logActivity.routes.ts,	VisitorLog, LogActivity,	Requests mutate -> auto	Protected CMS analytics/log

Logs	GA4 analytics, activity audit logs.	activityLogger middleware/service	news_views	activity log; public visitor tracking -> dashboard; GA4 service queries external API	routes; public visitor endpoint should remain rate limited.
URL Redirect	Public redirect handling and CMS redirect rule management.	urlRedirect.routes.ts	UrlRedirect	Public /redirect/* -> controller evaluates rule; CMS manages rules	Protected CMS routes with url_redirection permissions.
Public Web	Corporate website rendering and public API proxy routes.	web/app, web/lib/cmsApi.ts, formsApi.js, eventsApi.js	Reads backend models via API	Browser/server components -> backend API -> rendered pages	Next security headers; no server-side auth for public pages.
Standalone File Manager Service	Separate API for S3 upload/list/delete/signed URLs.	filemanager/src/app.ts, filemanager/src/routes/file.routes.ts	S3 objects, no shared DB model found	Client -> /api -> optional x-api-key -> S3 service	API key disabled if API_KEY unset; must be configured in production.

7. Database Design

Database implementation uses Prisma ORM with PostgreSQL datasource. The following model/table inventory is generated from `backend/prisma/schema.prisma`.

Prisma Model	Table	Function	Primary Key	Foreign Key / Relationship	Field Count
User	users	Stores CMS/admin user identity, status, MFA, lockout, and profile attributes.	id	Not Found in Current Codebase	32
RefreshToken	refresh_tokens	Stores hashed refresh token/session metadata for JWT rotation and logout.	id	userId -> User.id	7
PasswordResetToken	password_reset_tokens	Stores password reset token hashes and expiry metadata.	id	Not Found in Current Codebase	6
PasswordHistory	password_histories	Stores previous password hashes to prevent reuse.	id	userId -> User.id	5
Role	roles	Defines authorization roles.	id	Not Found in Current Codebase	10
Permission	permissions	Defines granular permission codes.	id	Not Found in Current Codebase	8
RolePermission	role_permissions	Join table between roles and permissions.	id	permissionId -> Permission.id; roleId -> Role.id	6
UserRole	user_roles	Join table between users and roles.	id	roleId -> Role.id; userId -> User.id	6
Setting	settings	Stores public and CMS configuration values.	id	Not Found in Current Codebase	12
Menu	menus	Stores public navigation/menu hierarchy.	id	parentId -> Menu.id	24
Page	pages	Stores CMS pages, SEO metadata, status, and publishing data.	id	createdById -> User.id; updatedById -> User.id	28
PageComponent	page_components	Stores ordered page builder components and component data.	id	pageId -> Page.id	9
Career	careers	Stores career vacancies and career content references.	id	Not Found in Current Codebase	18
Award	awards	Stores awards and award display metadata.	id	Not Found in Current Codebase	22
ManagementCategory	management_categories	Stores management profile categories.	id	Not Found in Current Codebase	12
Management	managements	Stores management/director profile records.	id	categoryId -> ManagementCategory.id	21
ContactSubmission	contact_submissions	Stores public contact-us submissions.	id	userId -> User.id	17
LogActivity	log_activities	Stores CMS activity/audit trail records.	id	userId -> User.id	14
VisitorLog	visitor_logs	Stores visitor tracking data for dashboard analytics.	id	Not Found in Current Codebase	7
UrlRedirect	url_redirects	Stores redirect rules for public URL forwarding.	id	Not Found in Current Codebase	9
Folder	folders	Stores logical folder hierarchy for file manager.	id	parentId -> Folder.id	12
File	files	Stores file metadata, storage key, and ownership.	id	createdById -> User.id; folderId -> Folder.id	26
ContactUs	contact_us	Stores CMS-managed contact-us	id	Not Found in Current Codebase	16

FormModule	form_modules	information/content. Stores dynamic form module metadata and runtime definition.	id	Not Found in Current Codebase	26
FormStep	form_steps	Stores form wizard steps.	id	formModuleId -> FormModule.id	13
FormField	form_fields	Stores dynamic form field definitions.	id	formModuleId -> FormModule.id; formStepId -> FormStep.id; parentFieldId -> FormField.id	27
FormFieldOption	form_field_options	Stores selectable options for dynamic form fields.	id	fieldId -> FormField.id	12
FormFieldRule	form_field_rules	Stores dynamic visibility/behavior rules for forms.	id	formModuleId -> FormModule.id; sourceFieldId -> FormField.id; targetFieldId -> FormField.id	14
FormResponseConfig	form_response_configs	Stores post-submit response/redirect configuration.	id	formModuleId -> FormModule.id	14
FormIntegrationConfig	form_integration_configs	Stores external dispatch/integration configuration for form submissions.	id	formModuleId -> FormModule.id	13
FormSubmission	form_submissions	Stores dynamic form submission header data.	id	formModuleId -> FormModule.id	35
FormSubmissionValue	form_submission_values	Stores flattened submission field values.	id	submissionId -> FormSubmission.id	10
FormSubmissionGroup	form_submission_groups	Stores repeated/grouped form submission sections.	id	submissionId -> FormSubmission.id	10
FormSubmissionGroupValue	form_submission_group_values	Stores values under grouped submission sections.	id	groupId -> FormSubmissionGroup.id	10
FormSubmissionFile	form_submission_files	Stores uploaded file references attached to form submissions.	id	fileId -> File.id; groupId -> FormSubmissionGroup.id; submissionId -> FormSubmission.id	19
FormSubmissionDispatchLog	form_submission_dispatch_logs	Stores dispatch attempts to external integrations.	id	integrationConfigId -> FormIntegrationConfig.id; submissionId -> FormSubmission.id	15
CareerContent	career_content	Stores public career/life-at-linknet content.	id	Not Found in Current Codebase	17
ReportType	report_types	Stores report type/category records.	id	Not Found in Current Codebase	13
ReportSection	report_sections	Stores report sections under report types.	id	type_id -> ReportType.id	12
news	news	Stores news articles and multilingual/SEO fields.	id	category_id -> news_categories.id; created_by_id -> User.id; updated_by_id -> User.id	36
news_categories	news_categories	Stores news categories.	id	Not Found in Current Codebase	13
news_highlights	news_highlights	Stores highlighted news mapping.	id	news_id -> news.id	8
news_tag_relations	news_tag_relations	Join table between news and tags.	id	news_id -> news.id; tag_id -> news_tags.id	6
news_tags	news_tags	Stores news tags.	id	Not Found in Current Codebase	7
news_views	news_views	Stores news view analytics.	id	news_id -> news.id	7
events	events	Stores event content and registration metadata.	id	Not Found in Current Codebase	29
event_news_relations	event_news_relations	Join table between events and related news.	id	event_id -> events.id; news_id -> news.id	8
event_registrations	event_registrations	Stores event registration header data.	id	event_id -> events.id	16
event_registration_participants	event_registration_participants	Stores participants for event registrations.	id	registration_id -> event_registrations.id	9
reports	reports	Stores report items/files.	id	type_id -> ReportType.id; section_id -> ReportSection.id	27
AnnouncementType	announcement_types	Stores announcement type/category records.	id	Not Found in Current Codebase	14
AnnouncementSection	announcement_sections	Stores announcement sections under types.	id	type_id -> AnnouncementType.id	16
announcements	announcements	Stores announcement items.	id	type_id -> AnnouncementType.id; section_id -> AnnouncementSection.id	19
ComponentVisibility	component_visibility	Stores visibility controls for	id	Not Found in Current Codebase	7

		frontend components.			
LabelGroup	label_groups	Stores label data-bank groups.	id	Not Found in Current Codebase	8
LabelNode	label_nodes	Stores hierarchical labels under groups.	id	groupId -> LabelGroup.id; parentId -> LabelNode.id	17
CookieConsent	cookie_consent	Stores public cookie consent decisions and metadata.	id	Not Found in Current Codebase	8

7.1 Prisma Enums

Enum	Values Found
ComponentVisibilityStatus	ACTIVE, INACTIVE
LabelStatus	ACTIVE, INACTIVE
UserStatus	ACTIVE, INACTIVE, SUSPENDED
SettingType	STRING, NUMBER, BOOLEAN, JSON, IMAGE, SELECT
MenuPosition	HEADER @map("header"), FOOTER @map("footer"), BOTH @map("both")
MenuType	LINK @map("link"), DROPDOWN @map("dropdown"), MEGA @map("mega")
PageTemplate	DEFAULT, FULL_WIDTH, LANDING
PageStatus	DRAFT, PUBLISHED
ContentStatus	DRAFT, PUBLISHED, ARCHIVED
EmploymentType	FULL_TIME, PART_TIME, CONTRACT, INTERNSHIP
CareerStatus	OPEN, CLOSED, DRAFT
ContactType	GENERAL, SUPPORT, SALES, PARTNERSHIP, COMPLAINT
SubmissionStatus	NEW, READ, REPLIED, CLOSED
InquiryType	BUSINESS, SUPPORT, CAREER, OTHERS
ContactStatus	NEW, READ
BusinessUnit	ENTERPRISE, FIBER, MEDIA
FormCategory	REGISTRATION, INQUIRY, PARTNERSHIP, RECOMMENDATION, EVENT
FormHandlingMode	SUBMISSION, ROUTING_ONLY
FormModuleStatus	DRAFT, ACTIVE, ARCHIVED
FormFieldType	TEXT, EMAIL, PHONE, NUMBER, TEXTAREA, SELECT, MULTI_SELECT, CHECKBOX, CHECKBOX_GROUP, RADIO, DATE, FILE, FILE_GROUP, ADDRESS_LOOKUP, REPEATER, HIDDEN
FormRuleType	SHOW, HIDE, REQUIRE, CLEAR, DISABLE, SYNC, DERIVE, LIMIT
FormResponseType	SUCCESS, INCOMPLETE, REDIRECT
FormIntegrationProvider	INTERNAL, CRM_WEB_TO_LEAD, CUSTOM, NOOP
FormDispatchMode	SYNC, ASYNC
FormSubmissionStatus	STORED, FAILED
FormFileStatus	PENDING, UPLOADED, LINKED, FAILED
FormDispatchStatus	PENDING, SUCCESS, FAILED, SKIPPED

7.2 Migration Summary

Migration	Summary
20251119062953_init	Prisma migration directory found in current codebase
20251208042446_authmigration	Prisma migration directory found in current codebase
20260119033028_add_contact_us_table	Prisma migration directory found in current codebase
20260119034132_add_contact_status_fields	Prisma migration directory found in current codebase
20260119150417_enhanced_log_activity	Prisma migration directory found in current codebase
20260202135203_rebuild_menus_table	Prisma migration directory found in current codebase
20260223033453_management_mysql_compat	Prisma migration directory found in current codebase
20260223_news_cms_mysql_compat	Prisma migration directory found in current codebase
20260223_report_module	Prisma migration directory found in current codebase
20260402125916_add_cookie_consent_table	Prisma migration directory found in current codebase
20260422023748_init	Prisma migration directory found in current codebase
20260427143000_add_events_module	Prisma migration directory found in current codebase
20260427190000_expand_event_cms	Prisma migration directory found in current codebase
20260428043000_add_form_modules	Prisma migration directory found in current codebase
20260428101500_collapse_form_submission_statuses	Prisma migration directory found in current codebase
20260430100000_add_page_metadata_and_submission_context	Prisma migration directory found in current codebase
20260501134500_add_award_top_logo	Prisma migration directory found in current codebase
20260502011000_add_news_category_intro_fields	Prisma migration directory found in current codebase
20260502021000_drop_news_category_intro_fields	Prisma migration directory found in current codebase

20260502033000 add news detail settings	Prisma migration directory found in current codebase
20260502170000 add news seo visibility	Prisma migration directory found in current codebase
20260503100000 add timezone setting	Prisma migration directory found in current codebase
20260503103000 make timezone setting public	Prisma migration directory found in current codebase
20260503120000 refactor settings grouped keys	Prisma migration directory found in current codebase
20260503123000 revise settings tabs fields	Prisma migration directory found in current codebase
20260503130000 bilingual settings values	Prisma migration directory found in current codebase
20260503131500 bilingual about content	Prisma migration directory found in current codebase
20260503133000 add cookies more info title	Prisma migration directory found in current codebase
20260503140000 add page layout visibility	Prisma migration directory found in current codebase
20260503141000 add page bilingual titles	Prisma migration directory found in current codebase
20260503150000 add label data bank	Prisma migration directory found in current codebase
20260503153000 backfill label values from legacy names	Prisma migration directory found in current codebase
20260503160000 add manual label id flag	Prisma migration directory found in current codebase
20260503163000 normalize label ids and span values	Prisma migration directory found in current codebase
20260503170000 fix label child full path	Prisma migration directory found in current codebase
20260503183000 revise enterprise form modules	Prisma migration directory found in current codebase
20260503193000 add omni channel form module	Prisma migration directory found in current codebase
20260503194000 separate form submission info	Prisma migration directory found in current codebase
20260503210000 fix omni channel zip field	Prisma migration directory found in current codebase
20260503213000 add page meta thumbnail	Prisma migration directory found in current codebase
20260504100000 add review status to form submissions	Prisma migration directory found in current codebase
20260506160000 add event multilingual fields	Prisma migration directory found in current codebase
career_content	Prisma migration directory found in current codebase
manual_refresh_token_rotation.sql	Manual SQL migration found
manual_security_controls_008_013.sql	Manual SQL migration found

7.3 ERD Explanation

ERD can be derived from Prisma relations. Core relationships found include User -> RefreshToken/PasswordHistory/LogActivity/File/Page/ContactSubmission, User <-> Role through UserRole, Role <-> Permission through RolePermission, Page -> PageComponent, Menu self-parent hierarchy, File -> Folder/User, dynamic form module -> steps/fields/options/rules/submissions/dispatch logs, report/announcement type -> section -> item, event -> registrations/participants, and news -> category/tags/highlights/views. A rendered ERD diagram file was Not Found in Current Codebase.

8. Authentication & Authorization

Area	Implementation Found
Login flow	POST /api/v1/auth/login validates email/password, applies business-hour and login rate limiting, checks user status/lockout/password expiry, verifies bcrypt hash, evaluates MFA, issues tokens, and sets cookies.
Session/JWT handling	Access JWT and refresh JWT generated by jwt.util; refresh token stored hashed in refresh_tokens with tokenId/sessionId; refresh endpoint rotates token in transaction.
MFA implementation	TOTP setup/enable/disable/status/verify implemented; MFA is conditional on MFA_ENABLED and user mfaEnabled.
Role & permission	Roles and permissions stored in DB with join tables; route-level requirePermission/checkPermission/authorize middleware used for CMS modules.
Middleware security	authMiddleware validates bearer/cookie token, user status, deletedAt, lockout, and refresh session; CSRF middleware protects cookie-auth unsafe API requests.
Password policy	Minimum 8 characters plus uppercase, lowercase, number, special character; disallows password equal to email/username in several flows; bcryptjs salt rounds found as 10.

9. API Documentation

9.1 API Conventions

Convention	Value
Backend base URL	API prefix defaults to /api/v1; root / and /health also exist.
Authentication	Bearer access token or HttpOnly auth_token cookie. Protected routes also require user to be active, unlocked, and not deleted.

CSRF	Unsafe /api methods require X-CSRF-Token when auth_token cookie is used; exemptions include login/logout/refresh/forgot/reset/MFA verify.
Content type	application/json for JSON endpoints; multipart/form-data for upload endpoints.
Common success shape	Controllers commonly return success/message/data/pagination. Exact field names vary by controller.
Common error shape	Global error handler returns success:false, error/message/statusCode/requestId and details when available.

9.2 Common Request and Response Examples

Auth Login Request

```
{
  "email": "admin@example.com",
  "password": "StrongPassword1!"
}
```

Auth Login Success Response

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "user": { "id": "uuid", "email": "admin@example.com", "roles": [] },
    "accessToken": "jwt-access-token",
    "refreshToken": "jwt-refresh-token",
    "csrfToken": "csrf-token"
  }
}
```

MFA Required Response

```
{
  "success": true,
  "mfaRequired": true,
  "tempToken": "temporary-token"
}
```

Generic Paginated Response

```
{
  "success": true,
  "data": [],
  "pagination": { "page": 1, "limit": 10, "total": 0, "totalPages": 0 }
}
```

Generic Error Response

```
{
  "success": false,
  "message": "Validation failed",
  "error": "Bad Request",
  "statusCode": 400,
  "requestId": "request-id",
  "details": []
}
```

9.3 Endpoint Inventory

Total endpoints inventoried from active mounts and route handlers: 339. The table below is generated from mounted route files in `backend/src/server.ts`, the standalone filemanager app, and Next.js route handlers under `web/app/api`.

Activity Log

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/log-activity	Get paginated activity logs with filters	Required; permission: log_activity.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getActivityLogs (backend/src/routes/logActivity.routes.ts:23)
Backend API	DELETE	/api/v1/cms/log-activity/:id	Soft delete activity log	Required; permission: log_activity.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteActivityLog (backend/src/routes/logActivity.routes.ts:51)
Backend API	GET	/api/v1/cms/log-activity/:id	Get activity log by ID with diff view	Required; permission: log_activity.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getActivityLogById (backend/src/routes/logActivity.routes.ts:44)
Backend API	POST	/api/v1/cms/log-activity/cleanup	Cleanup old logs (soft delete logs older than X days)	Required; permission: log_activity.delete	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	cleanupOldLogs (backend/src/routes/logActivity.routes.ts:58)
Backend API	GET	/api/v1/cms/log-activity/stats	Get activity log statistics	Required; permission: log_activity.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getActivityLogStats (backend/src/routes/logActivity.routes.ts:30)
Backend API	GET	/api/v1/cms/log-activity/user/:userId/timeline	Get user activity timeline	Required; permission: log_activity.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: userId; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getUserActivityTimeline (backend/src/routes/logActivity.routes.ts:37)

Analytics

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/analytics/combined	getCombinedAnalytics	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getCombinedAnalytics (backend/src/routes/analytics.routes.ts:27)
Backend API	GET	/api/v1/analytics/ga	getGoogleAnalytics	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getGoogleAnalytics (backend/src/routes/analytics.routes.ts:24)
Backend API	GET	/api/v1/analytics/ga/status	getGAStatus	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getGAStatus (backend/src/routes/analytics.routes.ts:25)
Backend API	GET	/api/v1/analytics/news	getNewsAnalytics	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getNewsAnalytics (backend/src/routes/analytics.routes.ts:26)

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Announcements

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/announcements/filter	announcementfilterAnnouncements	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	announcementController.filterAnnouncements (backend/src/routes/announcement.routes.ts:12)
Backend API	GET	/api/v1/announcements/section/id/items	announcementgetPublicSectionItems	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	announcementController.getPublicSectionItems (backend/src/routes/announcement.routes.ts:17)
Backend API	GET	/api/v1/cms/announcement-items	announcementgetAnnouncementItems	Required; permission: announcements.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementItems (backend/src/routes/announcement.routes.ts:202)
Backend API	POST	/api/v1/cms/announcement-items	announcementcreateAnnouncementItem	Required; permission: announcements.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.createAnnouncementItem (backend/src/routes/announcement.routes.ts:209)
Backend API	DELETE	/api/v1/cms/announcement-items/id	announcementdeleteAnnouncementItem	Required; permission: announcements.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.deleteAnnouncementItem (backend/src/routes/announcement.routes.ts:230)
Backend API	GET	/api/v1/cms/announcement-items/id	announcementgetAnnouncementItemById	Required; permission: announcements.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementItemById (backend/src/routes/announcement.routes.ts:216)
Backend API	PUT	/api/v1/cms/announcement-items/id	announcementupdateAnnouncementItem	Required; permission:	Content-Type: application/json	Path params: id;	Not Found in Current	Create/update/action success	400/422 validation,	announcementController.updateAnnouncementItem (backend/src/routes/announcement.routes.ts:223)

				announcements.update	n; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X- CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Codebase as centralized validator; controller/service may validate internally.	envelope; validation errors use 400/422 pattern.	404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	POST	/api/v1/cms/announcement-items/destroy-multiple	announcementdeleteMultipleAnnouncementItems	Required; permission: announcements.delete	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X- CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.deleteMultipleAnnouncementItems (backend/src/routes/announcement.routes.ts:188)
Backend API	GET	/api/v1/cms/announcement-items/stats	announcementgetAnnouncementItemStats	Required; permission: announcements.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementItemStats (backend/src/routes/announcement.routes.ts:174)
Backend API	POST	/api/v1/cms/announcement-items/toggle-status	announcementtoggleAnnouncementItemStatus	Required; permission: announcements.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X- CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.toggleAnnouncementItemStatus (backend/src/routes/announcement.routes.ts:181)
Backend API	POST	/api/v1/cms/announcement-items/update-order	announcementupdateAnnouncementItemsOrder	Required; permission: announcements.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X- CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.updateAnnouncementItemsOrder (backend/src/routes/announcement.routes.ts:195)
Backend API	GET	/api/v1/cms/announcement-sections	announcementgetAnnouncementSections	Required; permission: announcements.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementSections (backend/src/routes/announcement.routes.ts:135)
Backend	POST	/api/v1/cms/announcement-sections	announcementcreateAnnouncementSection	Required;	Content-Type:	Body:	Not Found in	Create/update/action	400/422	announcementController.createAnnouncementSection

d API				permission: announcements.create	application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	JSON payload for the target resource	Current Codebase as centralized validator; controller/service may validate internally.	on success envelope; validation errors use 400/422 pattern.	validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	(backend/src/routes/announcement.routes.ts:142)
Backend API	DELETE	/api/v1/cms/announcement-sections/:id	announcementdeleteAnnouncementSection	Required; permission: announcements.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.deleteAnnouncementSection (backend/src/routes/announcement.routes.ts:163)
Backend API	GET	/api/v1/cms/announcement-sections/:id	announcementgetAnnouncementSectionById	Required; permission: announcements.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementSectionById (backend/src/routes/announcement.routes.ts:149)
Backend API	PUT	/api/v1/cms/announcement-sections/:id	announcementupdateAnnouncementSection	Required; permission: announcements.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.updateAnnouncementSection (backend/src/routes/announcement.routes.ts:156)
Backend API	GET	/api/v1/cms/announcement-sections/:id/items	announcementgetAnnouncementSectionItems	Required; permission: announcements.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementSectionItems (backend/src/routes/announcement.routes.ts:121)
Backend API	POST	/api/v1/cms/announcement-sections/:id/items/update-order	announcementupdateSectionItemsOrder	Required; permission: announcements.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.updateSectionItemsOrder (backend/src/routes/announcement.routes.ts:128)

Backend API	POST	/api/v1/cms/announcement-sections/destroy-multiple	announcementdeleteMultipleAnnouncementSections	Required; permission: announcements.delete	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.deleteMultipleAnnouncementSections (backend/src/routes/announcement.routes.ts:114)
Backend API	GET	/api/v1/cms/announcement-sections/list	announcementgetAnnouncementSectionsList	Required; permission: announcements.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementSectionsList (backend/src/routes/announcement.routes.ts:100)
Backend API	POST	/api/v1/cms/announcement-sections/toggle-status	announcementtoggleAnnouncementSectionStatus	Required; permission: announcements.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.toggleAnnouncementSectionStatus (backend/src/routes/announcement.routes.ts:107)
Backend API	GET	/api/v1/cms/announcement-types	announcementgetAnnouncementTypes	Required; permission: announcements.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementTypes (backend/src/routes/announcement.routes.ts:61)
Backend API	POST	/api/v1/cms/announcement-types	announcementcreateAnnouncementType	Required; permission: announcements.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.createAnnouncementType (backend/src/routes/announcement.routes.ts:68)
Backend API	DELETE	/api/v1/cms/announcement-types/id	announcementdeleteAnnouncementType	Required; permission: announcements.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.deleteAnnouncementType (backend/src/routes/announcement.routes.ts:89)

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Backend API	GET	/api/v1/cms/announcement-types/id	announcementgetAnnouncementTypeById	Required; permission: announcements.read	Authorization: Bearer <accessToken > or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementTypeById (backend/src/routes/announcement.routes.ts:75)
Backend API	PUT	/api/v1/cms/announcement-types/id	announcementupdateAnnouncementType	Required; permission: announcements.update	Content-Type: application/json; Authorization: Bearer <accessToken > or HttpOnly auth_token cookie; X- CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.updateAnnouncementType (backend/src/routes/announcement.routes.ts:82)
Backend API	GET	/api/v1/cms/announcement-types/id/sections	announcementgetAnnouncementTypeSections	Required; permission: announcements.read	Authorization: Bearer <accessToken > or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementTypeSections (backend/src/routes/announcement.routes.ts:47)
Backend API	POST	/api/v1/cms/announcement-types/id/sections/update-order	announcementupdateSectionsOrder	Required; permission: announcements.update	Content-Type: application/json; Authorization: Bearer <accessToken > or HttpOnly auth_token cookie; X- CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.updateSectionsOrder (backend/src/routes/announcement.routes.ts:54)
Backend API	POST	/api/v1/cms/announcement-types/destroy-multiple	announcementdeleteMultipleAnnouncementTypes	Required; permission: announcements.delete	Content-Type: application/json; Authorization: Bearer <accessToken > or HttpOnly auth_token cookie; X- CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	announcementController.deleteMultipleAnnouncementTypes (backend/src/routes/announcement.routes.ts:40)
Backend API	GET	/api/v1/cms/announcement-types/list	announcementgetAnnouncementTypesList	Required; permission: announcements.read	Authorization: Bearer <accessToken > or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	announcementController.getAnnouncementTypesList (backend/src/routes/announcement.routes.ts:26)
Backend API	POST	/api/v1/cms/announcement-types/toggle-status	announcementtoggleAnnouncementTypeStatus	Required; permission: announcements.update	Content-Type: application/json;	Body: JSON payload for	Not Found in Current Codebase as	Create/update/action success envelope;	400/422 validation, 404 not	announcementController.toggleAnnouncementTypeStatus (backend/src/routes/announcement.routes.ts:33)

				ate	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	the target resource	centralized validator; controller/service may validate internally.	validation errors use 400/422 pattern.	found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
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Authentication

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	POST	/api/v1/auth/forgot-password	forgotPassword	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	forgotPasswordValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	forgotPassword (backend/src/routes/auth.routes.ts:52)
Backend API	POST	/api/v1/auth/login	login	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	loginValidation; login rate limiter	Auth success example with user, accessToken, refreshToken, csrfToken; or MFA required response.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	login (backend/src/routes/auth.routes.ts:43)
Backend API	POST	/api/v1/auth/logout	logout	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	logout (backend/src/routes/auth.routes.ts:46)
Backend API	POST	/api/v1/auth/logout-all	logoutAll	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	logoutAll (backend/src/routes/auth.routes.ts:84)
Backend API	GET	/api/v1/auth/me	getCurrentUser	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getCurrentUser (backend/src/routes/auth.routes.ts:81)
Backend API	POST	/api/v1/auth/mfa/disable	mfaDisable	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	mfaDisable (backend/src/routes/auth.routes.ts:71)
Backend API	POST	/api/v1/auth/mfa/enable	mfaEnable	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	mfaEnable (backend/src/routes/auth.routes.ts:68)
Backend API	POST	/api/v1/auth/mfa/setup	mfaSetup	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	mfaSetup (backend/src/routes/auth.routes.ts:65)
Backend API	GET	/api/v1/auth/mfa/status	mfaStatus	Required	Authorization:	No explicit body	Path/query validation not	Public success	400/422 validation,	mfaStatus

					Bearer <accessToken> or HttpOnly auth_token cookie	required	centralized unless handled in controller.	envelope with data object/array.	404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	(backend/src/routes/auth.routes.ts:74)
Backend API	POST	/api/v1/auth/mfa/verify	mfaVerify	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	mfaVerify (backend/src/routes/auth.routes.ts:62)
Backend API	POST	/api/v1/auth/refresh	refreshAccessToken	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	refreshTokenValidation	Token rotation success example.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	refreshAccessToken (backend/src/routes/auth.routes.ts:49)
Backend API	POST	/api/v1/auth/register	register	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	registerValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	register (backend/src/routes/auth.routes.ts:40)
Backend API	POST	/api/v1/auth/reset-password	resetPassword	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	resetPasswordValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	resetPassword (backend/src/routes/auth.routes.ts:55)

Awards

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/awards	awardgetActiveAwards	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: public query filters as supported by controller	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	awardController.getActiveAwards (backend/src/routes/award.routes.ts:9)
Backend API	GET	/api/v1/awards/by-year	awardgetAwardsByYear	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: public query filters as supported by controller	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	awardController.getAwardsByYear (backend/src/routes/award.routes.ts:10)
Backend API	GET	/api/v1/cms/awards	awardgetAwards	Required; permission: awards.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	awardController.getAwards (backend/src/routes/award.routes.ts:13)
Backend API	POST	/api/v1/cms/awards	awardcreateAward	Required; permission: awards.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	awardController.createAward (backend/src/routes/award.routes.ts:27)
Backend API	DELETE	/api/v1/cms/awards/:id	awarddeleteAward	Required; permission: awards.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	awardController.deleteAward (backend/src/routes/award.routes.ts:41)
Backend API	GET	/api/v1/cms/awards/:id	awardgetAwardById	Required; permission: awards.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	awardController.getAwardById (backend/src/routes/award.routes.ts:20)
Backend API	PUT	/api/v1/cms/awards/:id	awardupdateAward	Required; permission: awards.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500	awardController.updateAward (backend/src/routes/award.routes.ts:34)

					Token required when cookie auth is used				internal error	
Backend API	POST	/api/v1/cms/awards/update-order	awardupdateAwardsOrder	Required; permission: awards.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	awardController.updateAwardsOrder (backend/src/routes/award.routes.ts:48)

CMS Contact Us

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/contactus	Get all contact submissions with pagination	Required; permission: Permission.CONTACT_SUBMISSIONS_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	getContactSubmissionsValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getContactSubmissions (backend/src/routes/cms/contactus.routes.ts:32)
Backend API	DELETE	/api/v1/cms/contactus/:id	Delete contact submission by ID	Required; permission: Permission.CONTACT_SUBMISSIONS_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	deleteContactValidation	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteContactSubmission (backend/src/routes/cms/contactus.routes.ts:71)
Backend API	GET	/api/v1/cms/contactus/:id	Get contact submission detail (auto mark as read)	Required; permission: Permission.CONTACT_SUBMISSIONS_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	getContactByIdValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getContactSubmissionById (backend/src/routes/cms/contactus.routes.ts:45)
Backend API	PATCH	/api/v1/cms/contactus/:id/status	Update submission status (mark as read/unread)	Required; permission: Permission.CONTACT_SUBMISSIONS_REPLY	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	updateSubmissionStatusValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	updateSubmissionStatus (backend/src/routes/cms/contactus.routes.ts:58)
Backend API	POST	/api/v1/cms/contactus/destroy-multiple	Delete multiple contact submissions	Required; permission: Permission.CONTACT_SUBMISSIONS_DELETE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	bulkDeleteContactValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteMultipleSubmissions (backend/src/routes/cms/contactus.routes.ts:84)
Backend	GET	/api/v1/cms/contactus/export	Export	Required; permission:	Authorization:	Query:	Path/query validation not	Paginated/list or	400/422	exportContactSubmissions

d API			contact submissions to CSV	Permission.CONTACT_SUBMISSIONS_READ	Bearer <accessToken> or HttpOnly auth_token cookie	page, limit, search, sortBy, sortOrder, filters as supported by controller	centralized unless handled in controller.	detail success envelope; see common list/detail examples.	validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	(backend/src/routes/cms/contactus.routes.ts:97)
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CMS Form Modules

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/form-modules	getFormModules	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getFormModules (backend/src/modules/form-modules/formModuleCms.routes.ts:20)
Backend API	POST	/api/v1/cms/form-modules	createFormModule	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	createFormModule (backend/src/modules/form-modules/formModuleCms.routes.ts:22)
Backend API	DELETE	/api/v1/cms/form-modules/:id	deleteFormModule	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteFormModule (backend/src/modules/form-modules/formModuleCms.routes.ts:24)
Backend API	GET	/api/v1/cms/form-modules/:id	getFormModuleById	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getFormModuleById (backend/src/modules/form-modules/formModuleCms.routes.ts:21)
Backend API	PUT	/api/v1/cms/form-modules/:id	updateFormModule	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	updateFormModule (backend/src/modules/form-modules/formModuleCms.routes.ts:23)
Backend API	GET	/api/v1/cms/form-modules/:id/submissions	getFormModuleSubmissions	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getFormModuleSubmissions (backend/src/modules/form-modules/formModuleCms.routes.ts:25)
Backend API	GET	/api/v1/cms/form-modules/:id/submissions/:submissionId	getFormModuleSubmissionById	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id, submissionId; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getFormModuleSubmissionById (backend/src/modules/form-modules/formModuleCms.routes.ts:27)
Backend	POST	/api/v1/cms/form-	retryFormModuleSubmissionDispatches	Required	Content-Type:	Path params:	Not Found in	Create/update/action	400/422	retryFormModuleSubmissionDispatches

API		modules/id/submissions/:submissionId/retry-dispatch			application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	id, submissionId; Body: JSON payload for the target resource	Current Codebase as centralized validator; controller/service may validate internally.	success envelope; validation errors use 400/422 pattern.	validation, 404 not found, 401 unauthorized, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	(backend/src/modules/form-modules/formModuleCms.routes.ts:28)
Backend API	PATCH	/api/v1/cms/form-modules/id/submissions/:submissionId/status	updateFormModuleSubmissionStatus	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id, submissionId; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	updateFormModuleSubmissionStatus (backend/src/modules/form-modules/formModuleCms.routes.ts:29)
Backend API	GET	/api/v1/cms/form-modules/id/submissions/export	exportFormModuleSubmissions	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	exportFormModuleSubmissions (backend/src/modules/form-modules/formModuleCms.routes.ts:26)

CMS Pages

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/pages	PagegetPages	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	PageController.getPages (backend/src/routes/cms/page.routes.ts:9)
Backend API	POST	/api/v1/cms/pages	PagecreatePage	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	PageController.createPage (backend/src/routes/cms/page.routes.ts:13)
Backend API	DELETE	/api/v1/cms/pages/id	PagedeletePage	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	PageController.deletePage (backend/src/routes/cms/page.routes.ts:15)
Backend API	GET	/api/v1/cms/pages/id	PagegetPage	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	PageController.getPage (backend/src/routes/cms/page.routes.ts:12)
Backend API	PUT	/api/v1/cms/pages/id	PageupdatePage	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	PageController.updatePage (backend/src/routes/cms/page.routes.ts:14)
Backend API	PUT	/api/v1/cms/pages/id/components	PagesaveComponents	Required	Content-Type: application/json; Authorization: Bearer	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator;	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403	PageController.saveComponents (backend/src/routes/cms/page.routes.ts:16)

					<accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used		controller/service may validate internally.		forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	GET	/api/v1/cms/pages/id/history	PagegetPageHistory	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	PageController.getPageHistory (backend/src/routes/cms/page.routes.ts:11)
Backend API	GET	/api/v1/cms/pages/slug/check	PagecheckSlugAvailability	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	PageController.checkSlugAvailability (backend/src/routes/cms/page.routes.ts:10)

Careers

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/careers	careergetPublicCareers	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	careerListValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	careerController.getPublicCareers (backend/src/routes/career.routes.ts:41)
Backend API	GET	/api/v1/careers/slug	careergetBySlug	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: slug; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	careerSlugValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	careerController.getBySlug (backend/src/routes/career.routes.ts:54)
Backend API	GET	/api/v1/careers/filters	careergetFilterOptions	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	careerController.getFilterOptions (backend/src/routes/career.routes.ts:48)
Backend API	GET	/api/v1/cms/careers	careergetAdminCareers	Required; permission: careers.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	careerListValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	careerController.getAdminCareers (backend/src/routes/career.routes.ts:73)
Backend API	POST	/api/v1/cms/careers	careercreate	Required; permission: careers.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	createCareerValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	careerController.create (backend/src/routes/career.routes.ts:91)
Backend API	DELETE	/api/v1/cms/careers/id	careerdelete	Required; permission: careers.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	careerIdValidation	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	careerController.delete (backend/src/routes/career.routes.ts:109)
Backend API	GET	/api/v1/cms/careers/id	careergetById	Required; permission: careers.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	careerIdValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	careerController.getById (backend/src/routes/career.routes.ts:82)
Backend API	PUT	/api/v1/cms/careers/id	careerupdate	Required; permission: careers.update	Content-Type: application/json; Authorization:	Path params: id; Body: JSON payload for the	updateCareerValidation	Create/update/action success envelope; validation errors use	400/422 validation, 404 not found, 401	careerController.update (backend/src/routes/career.routes.ts:100)

					Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	target resource		400/422 pattern.	unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	POST	/api/v1/cms/careers/id/toggle-status	careertoggleStatus	Required; permission: careers.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	careerIdValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	careerController.toggleStatus (backend/src/routes/career.routes.ts:127)
Backend API	POST	/api/v1/cms/careers/bulk-delete	careerbulkDelete	Required; permission: careers.delete	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	bulkDeleteCareerValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	careerController.bulkDelete (backend/src/routes/career.routes.ts:118)
Backend API	GET	/api/v1/cms/careers/stats	careergetStats	Required; permission: careers.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	careerController.getStats (backend/src/routes/career.routes.ts:65)

Component Visibility

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/component-visibility	ComponentVisibilitygetAll	Required; permission: Permission.PAGES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	ComponentVisibilityController.getAll (backend/src/routes/componentVisibility.routes.ts:15)
Backend API	POST	/api/v1/cms/component-visibility	ComponentVisibilitycreate	Required; permission: Permission.PAGES_CREATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentVisibilityController.create (backend/src/routes/componentVisibility.routes.ts:45)
Backend API	DELETE	/api/v1/cms/component-visibility/:id	ComponentVisibilitydelete	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentVisibilityController.delete (backend/src/routes/componentVisibility.routes.ts:94)
Backend API	GET	/api/v1/cms/component-visibility/:id	ComponentVisibilitygetById	Required; permission: Permission.PAGES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	ComponentVisibilityController.getById (backend/src/routes/componentVisibility.routes.ts:35)
Backend API	PUT	/api/v1/cms/component-visibility/:id	ComponentVisibilityupdate	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer	Path params: id; Body: JSON payload for	Not Found in Current Codebase as centralized	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized,	ComponentVisibilityController.update (backend/src/routes/componentVisibility.routes.ts:74)

					<accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	the target resource	validator; controller/service may validate internally.		403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	PATCH	/api/v1/cms/component-visibility/id/toggle	ComponentVisibilitytoggleStatus	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentVisibilityController.toggleStatus (backend/src/routes/componentVisibility.routes.ts:83)
Backend API	POST	/api/v1/cms/component-visibility/bulk-toggle	ComponentVisibilitybulkToggle	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentVisibilityController.bulkToggle (backend/src/routes/componentVisibility.routes.ts:65)
Backend API	GET	/api/v1/cms/component-visibility/inactive-keys	ComponentVisibilitygetInactiveKeys	Required; permission: Permission.PAGES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	ComponentVisibilityController.getInactiveKeys (backend/src/routes/componentVisibility.routes.ts:26)
Backend API	POST	/api/v1/cms/component-visibility/sync	ComponentVisibilitiesyncFromRegistry	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentVisibilityController.syncFromRegistry (backend/src/routes/componentVisibility.routes.ts:55)

Contact Us

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/contact-us	getAllContactSubmissions	Required; permission: Permission.CONTACT_SUBMISSIONS_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getAllContactSubmissions (backend/src/routes/contact.routes.ts:25)
Backend API	DELETE	/api/v1/contact-us/id	deleteContactSubmission	Required; permission: Permission.CONTACT_SUBMISSIONS_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	deleteContactValidation	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteContactSubmission (backend/src/routes/contact.routes.ts:27)
Backend API	GET	/api/v1/contact-us/id	getContactSubmissionById	Required; permission: Permission.CONTACT_SUBMISSIONS_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id	getContactByIdValidation	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getContactSubmissionById (backend/src/routes/contact.routes.ts:26)
Backend API	POST	/api/v1/contact-us/submit	submitContactForm	Public / Not required	Content-Type: application/json	Body: JSON payload for	submitContactValidation	Create/update/action success envelope;	400/422 validation, 404	submitContactForm (backend/src/routes/contact.routes.ts:22)

						the target resource		validation errors use 400/422 pattern.	not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
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Cookie Consent

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/cookie-consents	cookieConsentgetConsents	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	cookieConsentController.getConsents (backend/src/routes/cookieConsent.routes.ts:33)
Backend API	DELETE	/api/v1/cms/cookie-consents/id	cookieConsentdeleteConsent	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	cookieConsentController.deleteConsent (backend/src/routes/cookieConsent.routes.ts:63)
Backend API	POST	/api/v1/cms/cookie-consents/destroy-multiple	cookieConsentdeleteMultiple	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	cookieConsentController.deleteMultiple (backend/src/routes/cookieConsent.routes.ts:73)
Backend API	GET	/api/v1/cms/cookie-consents/export	cookieConsentexportConsents	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	cookieConsentController.exportConsents (backend/src/routes/cookieConsent.routes.ts:53)
Backend API	GET	/api/v1/cms/cookie-consents/stats	cookieConsentgetStats	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	cookieConsentController.getStats (backend/src/routes/cookieConsent.routes.ts:43)
Backend API	POST	/api/v1/cookies/consent	cookieConsentrecordConsent	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	cookieConsentController.recordConsent (backend/src/routes/cookieConsent.routes.ts:20)

Dashboard

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/dashboard/content	getContentOverview	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getContentOverview (backend/src/routes/dashboard.routes.ts:19)
Backend API	GET	/api/v1/cms/dashboard/recent-activity	getRecentActivity	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getRecentActivity (backend/src/routes/dashboard.routes.ts:20)

Backend API	GET	/api/v1/cms/dashboard/visitors	getVisitorStats	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getVisitorStats (backend/src/routes/dashboard.routes.ts:17)
Backend API	GET	/api/v1/cms/dashboard/visitors/chart	getVisitorChartData	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getVisitorChartData (backend/src/routes/dashboard.routes.ts:18)
Backend API	POST	/api/v1/public/track-visit	trackVisit	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	trackVisit (backend/src/routes/dashboard.routes.ts:14)

Events

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/admin/events	eventgetEvents	Required; permission: events.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.getEvents (backend/src/routes/event.routes.ts:13)
Backend API	POST	/api/v1/admin/events	eventcreateEvent	Required; permission: events.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	eventController.createEvent (backend/src/routes/event.routes.ts:48)
Backend API	GET	/api/v1/admin/events-slug/check	eventcheckSlugAvailability	Required; permission: events.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.checkSlugAvailability (backend/src/routes/event.routes.ts:20)
Backend API	DELETE	/api/v1/admin/events/:id	eventdeleteEvent	Required; permission: events.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	eventController.deleteEvent (backend/src/routes/event.routes.ts:62)
Backend API	GET	/api/v1/admin/events/:id	eventgetEventById	Required; permission: events.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.getEventById (backend/src/routes/event.routes.ts:27)
Backend API	PUT	/api/v1/admin/events/:id	eventupdateEvent	Required; permission: events.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500	eventController.updateEvent (backend/src/routes/event.routes.ts:55)

Backend API	GET	/api/v1/admin/events/:id/registrations	eventgetEventRegistrations	Required; permission: events.read	is used Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.getEventRegistrations (backend/src/routes/event.routes.ts:34)
Backend API	GET	/api/v1/admin/events/:id/registrations/:registrationId	eventgetEventRegistrationById	Required; permission: events.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id, registrationId; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.getEventRegistrationById (backend/src/routes/event.routes.ts:41)
Backend API	GET	/api/v1/cms/events	eventgetEvents	Required; permission: events.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.getEvents (backend/src/routes/event.routes.ts:13)
Backend API	POST	/api/v1/cms/events	eventcreateEvent	Required; permission: events.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	eventController.createEvent (backend/src/routes/event.routes.ts:48)
Backend API	GET	/api/v1/cms/events-slug/check	eventcheckSlugAvailability	Required; permission: events.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.checkSlugAvailability (backend/src/routes/event.routes.ts:20)
Backend API	DELETE	/api/v1/cms/events/:id	eventdeleteEvent	Required; permission: events.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	eventController.deleteEvent (backend/src/routes/event.routes.ts:62)
Backend API	GET	/api/v1/cms/events/:id	eventgetEventById	Required; permission: events.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.getEventById (backend/src/routes/event.routes.ts:27)
Backend API	PUT	/api/v1/cms/events/:id	eventupdateEvent	Required; permission: events.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	eventController.updateEvent (backend/src/routes/event.routes.ts:55)
Backend API	GET	/api/v1/cms/events/:id/registrations	eventgetEventRegistrations	Required; permission: events.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	eventController.getEventRegistrations (backend/src/routes/event.routes.ts:34)
Backend	GET	/api/v1/cms/events/:id/registrations/:registrationId	eventgetEventRegistrationById	Required;	Authorization:	Path params:	Path/query	Paginated/list or	400/422	eventController.getEventRegistrationById

API				permission: events.read	Bearer <accessToken> or HttpOnly auth_token cookie	id, registrationId; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	validation not centralized unless handled in controller.	detail success envelope; see common list/detail examples.	validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	(backend/src/routes/event.routes.ts:41)
Backend API	GET	/api/v1/events	eventgetPublishedEvents	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	eventController.getPublishedEvents (backend/src/routes/event.routes.ts:8)
Backend API	GET	/api/v1/events/slug	eventgetPublishedEventBySlug	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: slug; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	eventController.getPublishedEventBySlug (backend/src/routes/event.routes.ts:10)
Backend API	POST	/api/v1/events/slug/registrations	eventcreateEventRegistration	Public / Not required	Content-Type: application/json	Path params: slug; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	eventController.createEventRegistration (backend/src/routes/event.routes.ts:9)

Example API

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/examples/async-error	failed	Public / Not required	Standard HTTP headers; Content- Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	failed (backend/src/routes/example.routes.ts:134)
Backend API	GET	/api/v1/examples/custom-validation	message	Public / Not required	Standard HTTP headers; Content- Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	message (backend/src/routes/example.routes.ts:64)
Backend API	GET	/api/v1/examples/database-error	timeout	Public / Not required	Standard HTTP headers; Content- Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	timeout (backend/src/routes/example.routes.ts:109)
Backend API	GET	/api/v1/examples/not-found/:id	found	Public / Not required	Standard HTTP headers; Content- Type when body is sent	Path params: id	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	found (backend/src/routes/example.routes.ts:78)
Backend API	GET	/api/v1/examples/success	response	Public / Not required	Standard HTTP headers; Content- Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	response (backend/src/routes/example.routes.ts:23)
Backend API	GET	/api/v1/examples/unauthorized	resource	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	resource (backend/src/routes/example.routes.ts:95)
Backend API	GET	/api/v1/examples/unexpected-error	happened	Public / Not required	Standard HTTP headers; Content- Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	happened (backend/src/routes/example.routes.ts:122)
Backend API	POST	/api/v1/examples/validate	req	Public / Not required	Content-Type: application/json	Body: JSON payload for the target resource	express-validator validateRequest	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	req (backend/src/routes/example.routes.ts:40)

File API

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/files	List files with optional prefix filter and pagination	Required; permission: Permission.FILES_READ	Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	listFiles (backend/src/routes/file.routes.ts:33)
Backend API	DELETE	/api/v1/files/:key	Delete a single file by cloud key	Required; permission: Permission.FILES_DELETE	Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: key	Path/query validation not centralized unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteFileByKey (backend/src/routes/file.routes.ts:48)
Backend API	POST	/api/v1/files/delete	Bulk delete files by cloud keys	Required; permission: Permission.FILES_DELETE	Content-Type: multipart/form-data or application/json as implemented; Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	bulkDeleteFiles (backend/src/routes/file.routes.ts:56)
Backend API	GET	/api/v1/files/info	Get single file metadata	Required; permission: Permission.FILES_READ	Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getFileInfo (backend/src/routes/file.routes.ts:41)

File Manager

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/filemanager/files	Get files with pagination and filters	Required; permission: Permission.FILES_READ	Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getFiles (backend/src/routes/filemanager.routes.ts:38)
Backend API	DELETE	/api/v1/filemanager/files/:id	Delete file	Required; permission: Permission.FILES_DELETE	Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteFile (backend/src/routes/filemanager.routes.ts:59)
Backend API	POST	/api/v1/filemanager/folder	Create new folder	Required; permission: Permission.FOLDERS_CREATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	createFolder (backend/src/routes/filemanager.routes.ts:52)
Backend API	GET	/api/v1/filemanager/folders	Get folder tree structure	Required; permission: Permission.FOLDERS_READ	Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getFolders (backend/src/routes/filemanager.routes.ts:45)
Backend API	POST	/api/v1/filemanager/move	Move files to another folder	Required; permission: Permission.FILES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly_auth_token cookie; X-CSRF-Token required when	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500	moveFiles (backend/src/routes/filemanager.routes.ts:66)

Backend API	GET	/api/v1/filemanager/search	Search files by name, type, tags	Required; permission: Permission.FILES_READ	cookie auth is used Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	internal error 400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	searchFiles (backend/src/routes/filemanager.routes.ts:73)
Backend API	POST	/api/v1/filemanager/upload	Upload files to cloud storage	Required; permission: Permission.FILES_CREATE	Content-Type: multipart/form-data or application/json as implemented; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: multipart file/files or JSON payload according to endpoint	multipart upload middleware	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	uploadFiles (backend/src/routes/filemanager.routes.ts:23)

File Manager V2

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/fm	List files with pagination & filtering	Required; permission: Permission.FILES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	listFilesV2 (backend/src/routes/filemanagerV2.routes.ts:46)
Backend API	DELETE	/api/v1/fm/:id	Delete file	Required; permission: Permission.FILES_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteFileV2 (backend/src/routes/filemanagerV2.routes.ts:60)
Backend API	GET	/api/v1/fm/:id	Get file details or download	Required; permission: Permission.FILES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getFileV2 (backend/src/routes/filemanagerV2.routes.ts:53)
Backend API	POST	/api/v1/fm/upload	Upload files via storage abstraction	Required; permission: Permission.FILES_CREATE	Content-Type: multipart/form-data or application/json as implemented; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: multipart file/files or JSON payload according to endpoint	multipart upload middleware; file scanner	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	uploadFileV2 (backend/src/routes/filemanagerV2.routes.ts:31)

Health

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/env-check	Environment and Azure Key Vault validation	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	envCheck (backend/src/routes/health.routes.ts:37)
Backend API	GET	/health	Basic health check - Kubernetes liveness probe	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	basicHealthCheck (backend/src/routes/health.routes.ts:23)
Backend API	GET	/health/detailed	Detailed health information for debugging	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	detailedHealthCheck (backend/src/routes/health.routes.ts:44)
Backend API	GET	/ready	Readiness check - Kubernetes readiness probe	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	readinessCheck (backend/src/routes/health.routes.ts:30)

Label Data Bank

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/labels	LabelDataBankgetGroups	Required; permission: Permission.PAGES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	LabelDataBankController.getGroups (backend/src/routes/labelDataBank.routes.ts:13)
Backend API	POST	/api/v1/cms/labels	LabelDataBankcreateGroup	Required; permission: Permission.PAGES_CREATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	LabelDataBankController.createGroup (backend/src/routes/labelDataBank.routes.ts:14)
Backend API	DELETE	/api/v1/cms/labels/:id	LabelDataBankdeleteGroup	Required; permission: Permission.PAGES_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	LabelDataBankController.deleteGroup (backend/src/routes/labelDataBank.routes.ts:16)
Backend API	PUT	/api/v1/cms/labels/:id	LabelDataBankupdateGroup	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	LabelDataBankController.updateGroup (backend/src/routes/labelDataBank.routes.ts:15)
Backend API	GET	/api/v1/cms/labels/:parent/tree	LabelDataBankgetTree	Required; permission: Permission.PAGES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: parent; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	LabelDataBankController.getTree (backend/src/routes/labelDataBank.routes.ts:18)
Backend API	POST	/api/v1/cms/labels/:parent/tree	LabelDataBankcreateLabel	Required; permission: Permission.PAGES_CREATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: parent; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	LabelDataBankController.createLabel (backend/src/routes/labelDataBank.routes.ts:19)
Backend API	DELETE	/api/v1/cms/labels/:parent/tree/:id	LabelDataBankdeleteLabel	Required; permission: Permission.PAGES_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: parent, id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	LabelDataBankController.deleteLabel (backend/src/routes/labelDataBank.routes.ts:22)
Backend API	PUT	/api/v1/cms/labels/:parent/tree/:id	LabelDataBankupdateLabel	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken>	Path params: parent, id; Body: JSON payload for the target	Not Found in Current Codebase as centralized validator;	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden,	LabelDataBankController.updateLabel (backend/src/routes/labelDataBank.routes.ts:20)

					or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	resource	controller/service may validate internally.		403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	PATCH	/api/v1/cms/labels/:parent/tree/:id/move	LabelDataBankmoveLabel	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: parent, id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	LabelDataBankController.moveLabel (backend/src/routes/labelDataBank.routes.ts:21)
Backend API	GET	/api/v1/public/labels	LabelDataBankgetPublicLabels	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: public query filters as supported by controller	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	LabelDataBankController.getPublicLabels (backend/src/routes/labelDataBank.routes.ts:9)

Linknet Enterprise Proxy

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/linknet-enterprise/cities	getEnterpriseCities	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getEnterpriseCities (backend/src/routes/linknetEnterprise.routes.ts:29)
Backend API	GET	/api/v1/linknet-enterprise/coverage	getEnterpriseCoverage	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	enterprise rate limiter	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getEnterpriseCoverage (backend/src/routes/linknetEnterprise.routes.ts:27)
Backend API	GET	/api/v1/linknet-enterprise/nearest	getEnterpriseNearest	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	enterprise rate limiter	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getEnterpriseNearest (backend/src/routes/linknetEnterprise.routes.ts:28)

Linknet Media Proxy

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/linknet-media	getLinknetMedia	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getLinknetMedia (backend/src/routes/linknetMedia.routes.ts:7)

Management

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/managements	Get all managements with pagination	Required; permission: management.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	managementController.getManagements (backend/src/routes/management.routes.ts:121)
Backend API	POST	/api/v1/cms/managements	Create new management	Required; permission: management.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500	managementController.createManagement (backend/src/routes/management.routes.ts:145)

Backend API	DELETE	/api/v1/cms/managements/id	Delete management	Required; permission: management.delete	auth is used Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	internal error 400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	managementController.deleteManagement (backend/src/routes/management.routes.ts:169)
Backend API	GET	/api/v1/cms/managements/id	Get single management by ID	Required; permission: management.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	managementController.getManagementById (backend/src/routes/management.routes.ts:133)
Backend API	PUT	/api/v1/cms/managements/id	Update management	Required; permission: management.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	managementController.updateManagement (backend/src/routes/management.routes.ts:157)
Backend API	POST	/api/v1/cms/managements/bulk-delete	Bulk delete managements	Required; permission: management.delete	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	managementController.bulkDeleteManagements (backend/src/routes/management.routes.ts:93)
Backend API	GET	/api/v1/cms/managements/categories	Get all management categories	Required; permission: management.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	managementController.getCategories (backend/src/routes/management.routes.ts:29)
Backend API	POST	/api/v1/cms/managements/categories	Create new category	Required; permission: management_categories.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	managementController.createCategory (backend/src/routes/management.routes.ts:53)
Backend API	DELETE	/api/v1/cms/managements/categories/id	Delete category	Required; permission: management_categories.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	managementController.deleteCategory (backend/src/routes/management.routes.ts:77)
Backend API	GET	/api/v1/cms/managements/categories/id	Get single category by ID	Required; permission: management.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	managementController.getCategoryById (backend/src/routes/management.routes.ts:41)
Backend API	PUT	/api/v1/cms/managements/categories/id	Update category	Required; permission: management_categories.update	Content-Type: application/json; Authorization:	Path params: id; Body: JSON	Not Found in Current Codebase as	Create/update/action success envelope; validation errors use	400/422 validation, 404 not found, 401	managementController.updateCategory (backend/src/routes/management.routes.ts:65)

					Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	payload for the target resource	centralized validator; controller/service may validate internally.	400/422 pattern.	unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	POST	/api/v1/cms/managements/categories/update-order	Update categories order (drag & drop)	Required; permission: management.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	managementController.updateCategoriesOrder (backend/src/routes/management.routes.ts:17)
Backend API	POST	/api/v1/cms/managements/update-order	Update managements data_order (drag & drop)	Required; permission: management.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	managementController.updateManagementsOrder (backend/src/routes/management.routes.ts:105)

Menu

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/menu	menugetMenus	Required; permission: Permission.MENU_MANAGEMENT_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	getMenusValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	menuController.getMenus (backend/src/routes/menu.routes.ts:26)
Backend API	POST	/api/v1/cms/menu	menucreateMenu	Required; permission: Permission.MENU_MANAGEMENT_CREATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	createMenuValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	menuController.createMenu (backend/src/routes/menu.routes.ts:51)
Backend API	DELETE	/api/v1/cms/menu/:id	menudeleteMenu	Required; permission: Permission.MENU_MANAGEMENT_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	deleteMenuValidation	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	menuController.deleteMenu (backend/src/routes/menu.routes.ts:69)
Backend API	GET	/api/v1/cms/menu/:id	menugetMenuById	Required; permission: Permission.MENU_MANAGEMENT_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported	getMenuByIdValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal	menuController.getMenuById (backend/src/routes/menu.routes.ts:42)

						by controller			error	
Backend API	PUT	/api/v1/cms/menu/:id	menuupdateMenu	Required; permission: Permission.MENU_MANAGEMENT_UPDATE	Content-Type: application/json ; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	updateMenuValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	menuController.updateMenu (backend/src/routes/menu.routes.ts:60)
Backend API	POST	/api/v1/cms/menu/destroy-multiple	menudeleteMultipleMenus	Required; permission: Permission.MENU_MANAGEMENT_DELETE	Content-Type: application/json ; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	deleteMultipleMenusValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	menuController.deleteMultipleMenus (backend/src/routes/menu.routes.ts:96)
Backend API	GET	/api/v1/cms/menu/flat	menugetAllMenusFlat	Required; permission: Permission.MENU_MANAGEMENT_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	menuController.getAllMenusFlat (backend/src/routes/menu.routes.ts:35)
Backend API	POST	/api/v1/cms/menu/toggle-status	menutoggleMenuStatus	Required; permission: Permission.MENU_MANAGEMENT_UPDATE	Content-Type: application/json ; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	toggleMenuStatusValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	menuController.toggleMenuStatus (backend/src/routes/menu.routes.ts:78)
Backend API	POST	/api/v1/cms/menu/update-order	menuupdateMenuOrder	Required; permission: Permission.MENU_MANAGEMENT_REORDER	Content-Type: application/json ; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	updateMenuOrderValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	menuController.updateMenuOrder (backend/src/routes/menu.routes.ts:87)
Backend API	GET	/api/v1/menu	menugetPublicMenus	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: public query filters as supported by controller	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	menuController.getPublicMenus (backend/src/routes/menu.routes.ts:22)
Backend API	GET	/api/v1/menu/position/:position	menugetMenusByPosition	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: position; Query: public query filters as supported by controller	getMenusByPositionValidation	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	menuController.getMenusByPosition (backend/src/routes/menu.routes.ts:23)

News

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/news	newsgetNews	Required; permission: news.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	newsController.getNews (backend/src/routes/news.routes.ts:35)
Backend API	POST	/api/v1/cms/news	newscreateNews	Required; permission: news.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsController.createNews (backend/src/routes/news.routes.ts:58)
Backend API	GET	/api/v1/cms/news-categories	newsCategorygetCategories	Required; permission: news.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	newsCategoryController.getCategories (backend/src/routes/news.routes.ts:134)
Backend API	POST	/api/v1/cms/news-categories	newsCategorycreateCategory	Required; permission: news.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsCategoryController.createCategory (backend/src/routes/news.routes.ts:173)
Backend API	DELETE	/api/v1/cms/news-categories/:id	newsCategorydeleteCategory	Required; permission: news.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsCategoryController.deleteCategory (backend/src/routes/news.routes.ts:197)
Backend API	GET	/api/v1/cms/news-categories/:id	newsCategorygetCategoryById	Required; permission: news.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	newsCategoryController.getCategoryById (backend/src/routes/news.routes.ts:165)
Backend API	PUT	/api/v1/cms/news-categories/:id	newsCategoryupdateCategory	Required; permission: news.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsCategoryController.updateCategory (backend/src/routes/news.routes.ts:181)
Backend API	PATCH	/api/v1/cms/news-categories/:id/status	newsCategorytoggleStatus	Required; permission: news.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsCategoryController.toggleStatus (backend/src/routes/news.routes.ts:189)
Backend	GET	/api/v1/cms/news-categories/active	newsCategorygetActiveCategories	Required	Authorization:	Query: page,	Path/query	Paginated/list or	400/422	newsCategoryController.getActiveCategories

API					Bearer <accessToken> or HttpOnly auth_token cookie	limit, search, sortBy, sortOrder, filters as supported by controller	validation not centralized unless handled in controller.	detail success envelope; see common list/detail examples.	validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	(backend/src/routes/news.routes.ts:142)
Backend API	DELETE	/api/v1/cms/news-categories/bulk	newsCategorybulkDeleteCategories	Required; permission: news.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body/query as implemented by controller	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsCategoryController.bulkDeleteCategories (backend/src/routes/news.routes.ts:157)
Backend API	PUT	/api/v1/cms/news-categories/reorder	newsCategoryupdateCategoryOrder	Required; permission: news.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsCategoryController.updateCategoryOrder (backend/src/routes/news.routes.ts:149)
Backend API	GET	/api/v1/cms/news-highlights	newsgetHighlights	Required; permission: news.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	newsController.getHighlights (backend/src/routes/news.routes.ts:84)
Backend API	POST	/api/v1/cms/news-highlights	newscreateHighlight	Required; permission: news.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsController.createHighlight (backend/src/routes/news.routes.ts:100)
Backend API	DELETE	/api/v1/cms/news-highlights/id	newsremoveHighlight	Required; permission: news.update	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsController.removeHighlight (backend/src/routes/news.routes.ts:124)
Backend API	GET	/api/v1/cms/news-highlights/available	newsgetAvailableForHighlight	Required; permission: news.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	newsController.getAvailableForHighlight (backend/src/routes/news.routes.ts:92)
Backend API	DELETE	/api/v1/cms/news-highlights/bulk	newsbulkRemoveHighlights	Required; permission: news.update	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body/query as implemented by controller	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsController.bulkRemoveHighlights (backend/src/routes/news.routes.ts:116)
Backend API	PUT	/api/v1/cms/news-highlights/reorder	newsreorderHighlights	Required; permission: news.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie	newsController.reorderHighlights (backend/src/routes/news.routes.ts:108)

					cookie; X-CSRF-Token required when cookie auth is used		internally.		token mismatch, 429 rate limit, 500 internal error	
Backend API	GET	/api/v1/cms/news-slug/check	newscheckSlugAvailability	Required; permission: news.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	newsController.checkSlugAvailability (backend/src/routes/news.routes.ts:42)
Backend API	DELETE	/api/v1/cms/news/:id	newsdeleteNews	Required; permission: news.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsController.deleteNews (backend/src/routes/news.routes.ts:74)
Backend API	GET	/api/v1/cms/news/:id	newsgetNewsById	Required; permission: news.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	newsController.getNewsById (backend/src/routes/news.routes.ts:50)
Backend API	PUT	/api/v1/cms/news/:id	newsupdateNews	Required; permission: news.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsController.updateNews (backend/src/routes/news.routes.ts:66)
Backend API	GET	/api/v1/news-categories	newsCategorygetActiveCategories	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	newsCategoryController.getActiveCategories (backend/src/routes/news.routes.ts:27)
Backend API	GET	/api/v1/news-categories/:slug	newsCategorygetCategoryBySlug	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: slug; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	newsCategoryController.getCategoryBySlug (backend/src/routes/news.routes.ts:30)
Backend API	POST	/api/v1/news-views/:newsId	newstrackNewsView	Public / Not required	Content-Type: application/json	Path params: newsId; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	newsController.trackNewsView (backend/src/routes/news.routes.ts:24)
Backend API	GET	/api/v1/public/news	newsgetActiveNews	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	newsController.getActiveNews (backend/src/routes/news.routes.ts:12)
Backend API	GET	/api/v1/public/news/:slug	newsgetNewsBySlug	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: slug; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	newsController.getNewsBySlug (backend/src/routes/news.routes.ts:18)
Backend API	GET	/api/v1/public/news/category/:categorySlug	newsgetNewsByCategorySlug	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: categorySlug; Query: page, limit, search, sortBy,	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	newsController.getNewsByCategorySlug (backend/src/routes/news.routes.ts:21)

						sortOrder, filters as supported by controller				
Backend API	GET	/api/v1/public/news/highlights	newsgetHighlightedNews	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	newsController.getHighlightedNews (backend/src/routes/news.routes.ts:15)

Page Components

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/pages/:pageId/components	ComponentgetPageComponents	Required; permission: Permission.PAGES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: pageId; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	ComponentController.getPageComponents (backend/src/routes/component.routes.ts:38)
Backend API	POST	/api/v1/cms/pages/:pageId/components	ComponentcreateComponent	Required; permission: Permission.PAGES_CREATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: pageId; Body: JSON payload for the target resource	createComponentValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentController.createComponent (backend/src/routes/component.routes.ts:49)
Backend API	POST	/api/v1/cms/pages/:pageId/components/reorder	ComponentreorderComponents	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: pageId; Body: JSON payload for the target resource	reorderComponentsValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentController.reorderComponents (backend/src/routes/component.routes.ts:62)
Backend API	GET	/api/v1/cms/pages/component-types	ComponentGetComponentTypes	Required; permission: Permission.PAGES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	ComponentController.GetComponentTypes (backend/src/routes/component.routes.ts:27)
Backend API	DELETE	/api/v1/cms/pages/components/:id	ComponentdeleteComponent	Required; permission: Permission.PAGES_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	deleteComponentValidation	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentController.deleteComponent (backend/src/routes/component.routes.ts:101)
Backend API	GET	/api/v1/cms/pages/components/:id	ComponentGetComponentById	Required; permission: Permission.PAGES_READ	Authorization: Bearer	Path params: id;	GetComponentByIdValidation	Paginated/list or detail success	400/422 validation,	ComponentController.GetComponentById (backend/src/routes/component.routes.ts:7)

					<accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller		envelope; see common list/detail examples.	404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	5)
Backend API	PUT	/api/v1/cms/pages/components/:id	ComponentupdateComponent	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	updateComponentValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentController.updateComponent (backend/src/routes/component.routes.ts:88)
Backend API	POST	/api/v1/cms/pages/components/:id/preview	ComponentpreviewComponent	Required; permission: Permission.PAGES_READ	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentController.previewComponent (backend/src/routes/component.routes.ts:127)
Backend API	POST	/api/v1/cms/pages/components/:id/toggle-visibility	ComponenttoggleVisibility	Required; permission: Permission.PAGES_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	GetComponentByIdValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	ComponentController.toggleVisibility (backend/src/routes/component.routes.ts:114)

Profile

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	DELETE	/api/v1/profile	deleteAccount	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body/query as implemented by controller	deleteAccountValidation	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteAccount (backend/src/routes/profile.routes.ts:42)
Backend API	GET	/api/v1/profile	getProfile	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getProfile (backend/src/routes/profile.routes.ts:27)
Backend API	PUT	/api/v1/profile	updateProfile	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	updateProfileValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	updateProfile (backend/src/routes/profile.routes.ts:30)
Backend API	DELETE	/api/v1/profile/avatar	deleteAvatar	Required	Authorization: Bearer <accessToken> or	Body/query as implemented by controller	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403	deleteAvatar (backend/src/routes/profile.routes.ts:36)

					HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used				forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	PUT	/api/v1/profile/avatar	updateAvatar	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	file scanner	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	updateAvatar (backend/src/routes/profile.routes.ts:33)
Backend API	PUT	/api/v1/profile/password	changePassword	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	changePasswordValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	changePassword (backend/src/routes/profile.routes.ts:39)

Public CMS Content

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/available-components	getAvailableComponents	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getAvailableComponents (backend/src/routes/public.routes.ts:16)
Backend API	POST	/api/v1/cms/pages/:id/revalidate	triggerRevalidation	Required	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	triggerRevalidation (backend/src/routes/public.routes.ts:36)
Backend API	GET	/api/v1/managements	managementgetActiveManagements	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	managementController.getActiveManagements (backend/src/routes/public.routes.ts:26)
Backend API	GET	/api/v1/managements/by-category	managementgetManagementsByCategory	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	managementController.getManagementsByCategory (backend/src/routes/public.routes.ts:30)
Backend API	GET	/api/v1/pages/:slug	getPublicPageBySlug	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: slug	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getPublicPageBySlug (backend/src/routes/public.routes.ts:23)
Backend API	GET	/api/v1/pages/preview/:slug	getPagePreview	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: slug	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getPagePreview (backend/src/routes/public.routes.ts:20)
Backend API	GET	/api/v1/pages/slugs	getPublishedSlugs	Public / Not required	Standard HTTP headers; Content-Type when body is sent	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getPublishedSlugs (backend/src/routes/public.routes.ts:19)

Public Form Modules

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	POST	/api/v1/form-enterprise/submit	submitEnterpriseFormModule	Public / Not required	Content-Type: application/json	Body: JSON payload for the	Not Found in Current Codebase	Create/update/action success envelope;	400/422 validation, 404	submitEnterpriseFormModule (backend/src/modules/form-

						target resource	as centralized validator; controller/service may validate internally.	validation errors use 400/422 pattern.	not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	modules/formModule.routes.ts:13)
Backend API	GET	/api/v1/forms/:businessUnit	getPublicFormModules	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: businessUnit; Query: public query filters as supported by controller	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getPublicFormModules (backend/src/modules/form-modules/formModule.routes.ts:14)
Backend API	GET	/api/v1/forms/:businessUnit/slug	getPublicFormModule	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: businessUnit, slug; Query: public query filters as supported by controller	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	getPublicFormModule (backend/src/modules/form-modules/formModule.routes.ts:15)
Backend API	POST	/api/v1/forms/:businessUnit/slug/files	uploadPublicFormFile	Public / Not required	Content-Type: multipart/form-data or application/json as implemented	Path params: businessUnit, slug; Body: multipart file/files or JSON payload according to endpoint	multipart upload middleware	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	uploadPublicFormFile (backend/src/modules/form-modules/formModule.routes.ts:17)
Backend API	POST	/api/v1/forms/:businessUnit/slug/submissions	submitPublicFormModule	Public / Not required	Content-Type: application/json	Path params: businessUnit, slug; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	submitPublicFormModule (backend/src/modules/form-modules/formModule.routes.ts:16)

Reports

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/report-items	Get items with pagination	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportItems (backend/src/routes/report.routes.ts:347)
Backend API	POST	/api/v1/cms/report-items	Create new item	Required; permission: reports.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.createReportItem (backend/src/routes/report.routes.ts:359)
Backend API	DELETE	/api/v1/cms/report-items/:id	Soft delete item	Required; permission: reports.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.deleteReportItem (backend/src/routes/report.routes.ts:395)
Backend API	GET	/api/v1/cms/report-items/:id	Get single item	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportItemById (backend/src/routes/report.routes.ts:371)
Backend API	PUT	/api/v1/cms/report-items/:id	Update item	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.updateReportItem (backend/src/routes/report.routes.ts:383)
Backend API	POST	/api/v1/cms/report-	Bulk soft delete	Required;	Content-Type:	Body: JSON array	Not Found in	Create/update/action	400/422 validation,	reportController.deleteMultipleReportItems

		items/destroy-multiple	items	permission: reports.delete	application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	of ids/keys	Current Codebase as centralized validator; controller/service may validate internally.	success envelope; validation errors use 400/422 pattern.	404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	(backend/src/routes/report.routes.ts:323)
Backend API	GET	/api/v1/cms/report-items/stats	Get report item stats	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportItemStats (backend/src/routes/report.routes.ts:299)
Backend API	POST	/api/v1/cms/report-items/toggle-status	Toggle item status	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.toggleReportItemStatus (backend/src/routes/report.routes.ts:311)
Backend API	POST	/api/v1/cms/report-items/update-order	Reorder items	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.updateReportItemsOrder (backend/src/routes/report.routes.ts:335)
Backend API	GET	/api/v1/cms/report-sections	Get sections with pagination	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportSections (backend/src/routes/report.routes.ts:235)
Backend API	POST	/api/v1/cms/report-sections	Create new section	Required; permission: reports.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.createReportSection (backend/src/routes/report.routes.ts:247)
Backend API	DELETE	/api/v1/cms/report-sections/id	Soft delete section	Required; permission: reports.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.deleteReportSection (backend/src/routes/report.routes.ts:283)
Backend API	GET	/api/v1/cms/report-sections/id	Get single section	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportSectionById (backend/src/routes/report.routes.ts:259)
Backend API	PUT	/api/v1/cms/report-sections/id	Update section	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.updateReportSection (backend/src/routes/report.routes.ts:271)
Backend API	GET	/api/v1/cms/report-sections/id/items	Get items for a section	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportSectionItems (backend/src/routes/report.routes.ts:211)
Backend API	POST	/api/v1/cms/report-sections/id/items/update-order	Reorder items within a section	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator;	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403	reportController.updateSectionItemsOrder (backend/src/routes/report.routes.ts:223)

					<accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used		controller/service may validate internally.		CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	POST	/api/v1/cms/report-sections/destroy-multiple	Bulk soft delete sections	Required; permission: reports.delete	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.deleteMultipleReportSections (backend/src/routes/report.routes.ts:199)
Backend API	GET	/api/v1/cms/report-sections/list	Get active sections for dropdown	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportSectionsList (backend/src/routes/report.routes.ts:175)
Backend API	POST	/api/v1/cms/report-sections/toggle-status	Toggle section status	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.toggleReportSectionStatus (backend/src/routes/report.routes.ts:187)
Backend API	GET	/api/v1/cms/report-types	Get report types with pagination	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportTypes (backend/src/routes/report.routes.ts:111)
Backend API	POST	/api/v1/cms/report-types	Create new report type	Required; permission: reports.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.createReportType (backend/src/routes/report.routes.ts:123)
Backend API	DELETE	/api/v1/cms/report-types/id	Soft delete report type	Required; permission: reports.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.deleteReportType (backend/src/routes/report.routes.ts:159)
Backend API	GET	/api/v1/cms/report-types/id	Get single report type	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportTypeById (backend/src/routes/report.routes.ts:135)
Backend API	PUT	/api/v1/cms/report-types/id	Update report type	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.updateReportType (backend/src/routes/report.routes.ts:147)
Backend API	GET	/api/v1/cms/report-types/id/sections	Get sections for a report type	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportTypeSections (backend/src/routes/report.routes.ts:87)
Backend API	POST	/api/v1/cms/report-types/id/sections/update-order	Reorder sections within a report type	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie;	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500	reportController.updateSectionsOrder (backend/src/routes/report.routes.ts:99)

					X-CSRF-Token required when cookie auth is used				internal error	
Backend API	POST	/api/v1/cms/report-types/destroy-multiple	Bulk soft delete report types	Required; permission: reports.delete	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.deleteMultipleReportTypes (backend/src/routes/report.routes.ts:75)
Backend API	GET	/api/v1/cms/report-types/list	Get active report types for dropdown	Required; permission: reports.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	reportController.getReportTypesList (backend/src/routes/report.routes.ts:51)
Backend API	POST	/api/v1/cms/report-types/toggle-status	Toggle report type status	Required; permission: reports.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	reportController.toggleReportTypeStatus (backend/src/routes/report.routes.ts:63)
Backend API	GET	/api/v1/reports/filter	Filter reports for public frontend (AJAX)	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	reportController.filterReports (backend/src/routes/report.routes.ts:17)
Backend API	GET	/api/v1/reports/section/id/items	Get items for a section (public modal)	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	reportController.getPublicSectionItems (backend/src/routes/report.routes.ts:37)
Backend API	GET	/api/v1/reports/years	Get available years for filter dropdown	Public / Not required	Standard HTTP headers; Content-Type when body is sent	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	reportController.getReportYears (backend/src/routes/report.routes.ts:27)

Role & Permission

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/roles	Get all roles with permissions	Required; permission: Permission.ROLE_MANAGEMENT_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getRoles (backend/src/routes/role.routes.ts:22)
Backend API	POST	/api/v1/cms/roles	Create new role	Required; permission: Permission.ROLE_MANAGEMENT_CREATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	createRole (backend/src/routes/role.routes.ts:44)
Backend API	DELETE	/api/v1/cms/roles/id	Delete role (blocked if is_system=true)	Required; permission: Permission.ROLE_MANAGEMENT_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteRole (backend/src/routes/role.routes.ts:58)
Backend API	GET	/api/v1/cms/roles/id	Get single role by ID	Required; permission: Permission.ROLE_MANAGEMENT_READ	Authorization: Bearer <accessToken> or	Path params: id; Query: page, limit, search,	Path/query validation not centralized unless	Paginated/list or detail success envelope; see	400/422 validation, 404 not found, 401	getRoleById (backend/src/routes/role.routes.ts:37)

					HttpOnly auth_token cookie	sortBy, sortOrder, filters as supported by controller	handled in controller.	common list/detail examples.	unauthorized, 403 forbidden, 429 rate limit, 500 internal error	
Backend API	PUT	/api/v1/cms/roles/:id	Update role (including permissions)	Required; permission: Permission.ROLE_MANAGEMENT_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	updateRole (backend/src/routes/role.routes.ts:51)
Backend API	GET	/api/v1/cms/roles/permissions	Get all permissions grouped by module	Required; permission: Permission.ROLE_MANAGEMENT_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getPermissions (backend/src/routes/role.routes.ts:30)

Root

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/	Root API server status endpoint	Public / Not required	Standard HTTP headers	No explicit body required	Not Found in Current Codebase	Status JSON with message, status, health/api endpoints.	500 internal error	inline server.ts handler (backend/src/server.ts)
Backend API	GET	/api/v1	API prefix status endpoint	Public / Not required	Standard HTTP headers	No explicit body required	Not Found in Current Codebase	API version/status JSON.	500 internal error	inline server.ts handler (backend/src/server.ts)

Settings

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/settings	SettingsgetAllSettings	Required; permission: Permission.SETTINGS_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	SettingsController.getAllSettings (backend/src/routes/settings.routes.ts:21)
Backend API	POST	/api/v1/cms/settings	SettingscreateSetting	Required; permission: Permission.SETTINGS_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	SettingsController.createSetting (backend/src/routes/settings.routes.ts:45)
Backend API	DELETE	/api/v1/cms/settings/:id	SettingsdeleteSetting	Required; permission: Permission.SETTINGS_UPDATE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	SettingsController.deleteSetting (backend/src/routes/settings.routes.ts:69)
Backend API	PUT	/api/v1/cms/settings/:id	SettingsupdateSetting	Required; permission: Permission.SETTINGS_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	SettingsController.updateSetting (backend/src/routes/settings.routes.ts:53)
Backend API	GET	/api/v1/cms/settings/:key	SettingsgetSettingByKey	Required; permission: Permission.SETTINGS_READ	Authorization: Bearer <accessToken> or	Path params: key; Query: page, limit,	Path/query validation not centralized unless	Paginated/list or detail success envelope; see	400/422 validation, 404 not found, 401	SettingsController.getSettingByKey (backend/src/routes/settings.routes.ts:37)

					HttpOnly auth_token cookie	search, sortBy, sortOrder, filters as supported by controller	handled in controller.	common list/detail examples.	unauthorized, 403 forbidden, 429 rate limit, 500 internal error	
Backend API	POST	/api/v1/cms/settings/clear-cache	SettingsclearCache	Required; permission: Permission.SETTINGS_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	SettingsController.clearCache (backend/src/routes/settings.routes.ts:77)
Backend API	GET	/api/v1/cms/settings/groups	SettingsgetGroups	Required; permission: Permission.SETTINGS_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	SettingsController.getGroups (backend/src/routes/settings.routes.ts:29)
Backend API	POST	/api/v1/cms/settings/update-group	SettingsupdateGroupSettings	Required; permission: Permission.SETTINGS_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	SettingsController.updateGroupSettings (backend/src/routes/settings.routes.ts:61)
Backend API	GET	/api/v1/settings/public	SettingsgetPublicSettings	Public / Not required	Standard HTTP headers; Content- Type when body is sent	Query: public query filters as supported by controller	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	SettingsController.getPublicSettings (backend/src/routes/settings.routes.ts:14)

Standalone File Manager Service

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Standalone File Manager	DELETE	/api/file	filedeleteFile	API key required when API_KEY env is configured; otherwise public in current implementation	x-api-key: <API_KEY> when configured; multipart/form-data for upload; application/json for bulk delete	Body/query as implemented by controller	Path/query validation unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	fileController.deleteFile (filemanager/src/routes/file.routes.ts:25)
Standalone File Manager	DELETE	/api/files	filebulkDeleteFiles	API key required when API_KEY env is configured; otherwise public in current implementation	x-api-key: <API_KEY> when configured; multipart/form-data for upload; application/json for bulk delete	Body/query as implemented by controller	Path/query validation unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	fileController.bulkDeleteFiles (filemanager/src/routes/file.routes.ts:33)
Standalone File Manager	GET	/api/files	filelistFiles	API key required when API_KEY env is configured; otherwise public in current implementation	x-api-key: <API_KEY> when configured; multipart/form-data for upload; application/json for bulk delete	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	fileController.listFiles (filemanager/src/routes/file.routes.ts:19)
Standalone File Manager	GET	/api/signed-url	filegetSignedUrl	API key required when API_KEY env is configured; otherwise public in current implementation	x-api-key: <API_KEY> when configured; multipart/form-data for upload; application/json for bulk delete	No explicit body required	Path/query validation unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 429 rate limit, 500 internal error	fileController.getSignedUrl (filemanager/src/routes/file.routes.ts:39)
Standalone File Manager	POST	/api/upload	fileuploadFile	API key required when API_KEY env is configured; otherwise public in current implementation	x-api-key: <API_KEY> when configured; multipart/form-data for upload; application/json for bulk delete	Body: multipart file/files or JSON payload according to endpoint	multipart upload middleware	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	fileController.uploadFile (filemanager/src/routes/file.routes.ts:13)

URL Redirect

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/url-redirects	Get all URL redirects with pagination	Required; permission: url_redirection.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getUrlRedirects (backend/src/routes/urlRedirect.routes.ts:40)
Backend API	POST	/api/v1/cms/url-redirects	Create new URL redirect	Required; permission: url_redirection.create	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	createUrlRedirect (backend/src/routes/urlRedirect.routes.ts:54)
Backend API	DELETE	/api/v1/cms/url-redirects/id	Delete URL redirect	Required; permission: url_redirection.delete	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id	Path/query validation not centralized unless handled in controller.	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteUrlRedirect (backend/src/routes/urlRedirect.routes.ts:68)
Backend API	GET	/api/v1/cms/url-redirects/id	Get URL redirect by ID	Required; permission: url_redirection.read	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	Path/query validation not centralized unless handled in controller.	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getUrlRedirectById (backend/src/routes/urlRedirect.routes.ts:47)
Backend API	PUT	/api/v1/cms/url-redirects/id	Update URL redirect	Required; permission: url_redirection.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	updateUrlRedirect (backend/src/routes/urlRedirect.routes.ts:61)
Backend API	PATCH	/api/v1/cms/url-redirects/id/toggle	Toggle URL redirect active status	Required; permission: url_redirection.update	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	toggleUrlRedirectStatus (backend/src/routes/urlRedirect.routes.ts:82)
Backend API	POST	/api/v1/cms/url-redirects/bulk-delete	Bulk delete URL redirects	Required; permission: url_redirection.delete	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	bulkDeleteUrlRedirects (backend/src/routes/urlRedirect.routes.ts:75)
Backend API	GET	/api/v1/redirect/*	Handle URL redirect (public)	Required	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Public success envelope with data object/array.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	handleRedirect (backend/src/routes/urlRedirect.routes.ts:26)

Upload

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	POST	/api/v1/upload	Upload single file via backend to cloud storage	Required; permission: Permission.FILES_CREATE	Content-Type: multipart/form-data or application/json as implemented; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: multipart file/files or JSON payload according to endpoint	multipart upload middleware; file scanner	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	uploadSingleFile (backend/src/routes/upload.routes.ts:33)
Backend API	POST	/api/v1/upload/multiple	Upload multiple files via backend to cloud storage	Required; permission: Permission.FILES_CREATE	Content-Type: multipart/form-data or application/json	Body: multipart file/files or JSON payload according	multipart upload middleware; file scanner	Upload/file result example with file metadata,	400/422 validation, 404 not found, 401 unauthorized, 403	uploadMultipleFiles (backend/src/routes/upload.routes.ts:48)

					as implemented; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	to endpoint		URL/key, and status.	forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	POST	/api/v1/upload/presigned	Get presigned URL for direct client-to- S3 upload	Required; permission: Permission.FILES_CREATE	Content-Type: multipart/form-data or application/json as implemented; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Body: multipart file/files or JSON payload according to endpoint	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	getPresignedUploadUrl (backend/src/routes/upload.routes.ts:64)
Backend API	GET	/api/v1/upload/presigned- get	Get presigned GET URL for private file access	Required; permission: Permission.FILES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation not centralized unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getPresignedGetUrl (backend/src/routes/upload.routes.ts:77)
Backend API	GET	/api/v1/upload/status	Get storage driver configuration status	Required; permission: Permission.FILES_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	No explicit body required	Path/query validation unless handled in controller.	Upload/file result example with file metadata, URL/key, and status.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getUploadStatus (backend/src/routes/upload.routes.ts:89)

User Management

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Backend API	GET	/api/v1/cms/users	Get paginated list of users with filters	Required; permission: Permission.USERS_MANAGEMENT_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	getUsersValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getUsers (backend/src/routes/user.routes.ts:35)
Backend API	POST	/api/v1/cms/users	Create new user with roles	Required; permission: Permission.USERS_MANAGEMENT_CREATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Body: JSON payload for the target resource	createUserValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	createUser (backend/src/routes/user.routes.ts:63)
Backend API	DELETE	/api/v1/cms/users/:id	Delete user (soft delete)	Required; permission: Permission.USERS_MANAGEMENT_DELETE	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF- Token required when cookie auth is used	Path params: id	deleteUserValidation	Delete success envelope; 404 if target not found.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	deleteUser (backend/src/routes/user.routes.ts:91)
Backend API	GET	/api/v1/cms/users/:id	Get user by ID with detailed information	Required; permission: Permission.USERS_MANAGEMENT_READ	Authorization: Bearer <accessToken> or HttpOnly auth_token cookie	Path params: id; Query: page, limit, search, sortBy, sortOrder, filters as supported by controller	getUserByIdValidation	Paginated/list or detail success envelope; see common list/detail examples.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 429 rate limit, 500 internal error	getUserById (backend/src/routes/user.routes.ts:49)
Backend API	PUT	/api/v1/cms/users/:id	Update user information and roles	Required; permission: Permission.USERS_MANAGEMENT_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly	Path params: id; Body: JSON payload for the target resource	updateUserValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF	updateUser (backend/src/routes/user.routes.ts:77)

					auth_token cookie; X-CSRF-Token required when cookie auth is used				when cookie token mismatch, 429 rate limit, 500 internal error	
Backend API	POST	/api/v1/cms/users/:id/force-password-change	MBSS2.0-010: Force user to change password on next login	Required; permission: Permission.USERS_MANAGEMENT_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	forcePasswordChange (backend/src/routes/user.routes.ts:145)
Backend API	POST	/api/v1/cms/users/:id/toggle-status	Toggle user status (ACTIVE <-> INACTIVE)	Required; permission: Permission.USERS_MANAGEMENT_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	toggleUserStatusValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	toggleUserStatus (backend/src/routes/user.routes.ts:105)
Backend API	POST	/api/v1/cms/users/:id/unlock	MBSS2.0-008: Unlock a locked user account (admin only)	Required; permission: Permission.USERS_MANAGEMENT_UPDATE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Path params: id; Body: JSON payload for the target resource	Not Found in Current Codebase as centralized validator; controller/service may validate internally.	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	unlockUserAccount (backend/src/routes/user.routes.ts:133)
Backend API	POST	/api/v1/cms/users/bulk-delete	Bulk delete users	Required; permission: Permission.USERS_MANAGEMENT_DELETE	Content-Type: application/json; Authorization: Bearer <accessToken> or HttpOnly auth_token cookie; X-CSRF-Token required when cookie auth is used	Body: JSON array of ids/keys	bulkDeleteUsersValidation	Create/update/action success envelope; validation errors use 400/422 pattern.	400/422 validation, 404 not found, 401 unauthorized, 403 forbidden, 403 CSRF when cookie token mismatch, 429 rate limit, 500 internal error	bulkDeleteUsers (backend/src/routes/user.routes.ts:119)

Web API Routes

Service	Method	Endpoint URL	Description	Authentication	Request Headers	Request Parameters / Body	Validation Rules	Response Example	Error Responses	Related Controller/Function
Public Web Next.js	POST	/api/form-enterprise/submit	Server-side proxy for enterprise form submission to backend API	Public / Not required	Standard HTTP headers; Content-Type: application/json for POST	Body: JSON payload for the target resource	Implemented inline in route handler; query/body checks vary by route	NextResponse JSON from backend/proxy handler	400 invalid request, 502 upstream error, 500 internal error	route.ts exported handler (web/app/api/form-enterprise/submit/route.ts)
Public Web Next.js	GET	/api/stock/historical	Yahoo Finance market data proxy	Public / Not required	Standard HTTP headers; Content-Type: application/json for POST	No explicit body required	Implemented inline in route handler; query/body checks vary by route	NextResponse JSON from backend/proxy handler	400 invalid request, 502 upstream error, 500 internal error	route.ts exported handler (web/app/api/stock/historical/route.ts)
Public Web Next.js	GET	/api/stock/quote	Yahoo Finance market data proxy	Public / Not required	Standard HTTP headers; Content-Type: application/json for POST	No explicit body required	Implemented inline in route handler; query/body checks vary by route	NextResponse JSON from backend/proxy handler	400 invalid request, 502 upstream error, 500 internal error	route.ts exported handler (web/app/api/stock/quote/route.ts)

10. Security Implementation

Security Area	Implementation Found
Authentication	JWT access/refresh tokens, HttpOnly cookies, session validation against refresh_tokens, optional bearer token.
Authorization	RBAC permission middleware on CMS endpoints; roles/permissions tables.
Password hashing	bcryptjs with salt rounds found as 10; password history and reset tokens stored as hashes.
CSRF protection	Custom CSRF middleware on /api for cookie-auth unsafe methods using X-CSRF-Token and csrf_token cookie.
XSS prevention	React default escaping, express-validator .escape in profile fields, Helmet/custom CSP. Dedicated HTML sanitizer not consistently found.
SQL Injection prevention	Prisma ORM used for database access; raw SQL usage should be reviewed separately if introduced.

File upload validation	MIME/extension/category/size checks; multer memory storage; scanUploadedFiles with ClamAV if available.
Rate limiting	General API limiter, login limiter, auth/strict/public limiters, enterprise proxy limiter. Development may disable global limiter by default.
Secure headers	Helmet plus custom headers: CSP default-src none, frame deny, nosniff, HSTS, permissions policy, no-store for sensitive routes, X-Robots-Tag for CMS.
Logging & monitoring	HTTP logger, request id, activity logger, log_activities persistence, GA4 analytics service. External APM Not Found in Current Codebase.
Environment variables	Startup validation for required envs; production JWT secret length validation; .env.example files found.
Cloud security integration	Azure Key Vault service and Kubernetes CSI manifests found; S3 SDK default credential chain/IRSA-friendly code found.
Storage security	Presigned URL endpoints for S3; file access authorization handled by route RBAC. Fine-grained object ACL policy Not Found in Current Codebase.

11. Configuration Management

Configuration Area	Implementation Found
Environment variables	Backend config centralizes app, CORS, JWT, rateLimit, upload; multiple .env.example files found.
Config files	backend/src/config/index.ts, database.ts, redis.ts; frontend/web next.config; Dockerfiles; Kubernetes YAML.
Secrets handling	Local .env files exist; Azure Key Vault service can fetch secrets with cache; GitHub Trivy secret scan workflow exists.
Storage configuration	UPLOAD_DIR, STORAGE_DRIVER/TYPE, AWS_S3 *, AZURE_STORAGE * variables.
External services	GA4, Google service account, Enterprise proxy credentials, Redis, SMTP/email-related code, Yahoo Finance library.

11.1 Environment Variables Summary

Scope	Source File	Keys Found
Backend example	backend/.env.example	NODE_ENV, PORT, HOST, API_PREFIX, API_VERSION, BASE_URL, DATABASE_URL, JWT_ACCESS_SECRET, JWT_ACCESS_EXPIRE, JWT_REFRESH_SECRET, JWT_REFRESH_EXPIRE, CORS_ORIGIN, CORS_CREDENTIALS, FRONTEND_URL, RATE_LIMIT_WINDOW_MS, RATE_LIMIT_MAX_REQUESTS, RATE_LIMIT_ENABLED, FORCE_HTTPS, FORCE_HTTPS_ALLOWED_HOSTS, ALLOWED_REDIRECT_HOSTS, AUTH_COOKIE_DOMAIN, STORAGE_DRIVER, MAX_FILE_SIZE, UPLOAD_DIR, AWS_REGION, AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY, AWS_S3_BUCKET, AWS_S3_PUBLIC_URL, AWS_S3_ENDPOINT, AZURE_STORAGE_CONNECTION_STRING, AZURE_STORAGE_ACCOUNT_NAME, AZURE_STORAGE_CONTAINER_NAME, AZURE_KEY_VAULT_URL, AZURE_TENANT_ID, AZURE_CLIENT_ID, AZURE_CLIENT_SECRET, MFA_ENABLED, MFA_APP_NAME, LOG_LEVEL, LOG_DIR, ENABLE_FILE_LOGS, REDIS_ENABLED, REDIS_HOST, REDIS_PORT, REDIS_PASSWORD, REDIS_DB, GA4_PROPERTY_ID, GOOGLE_SERVICE_ACCOUNT_JSON, SETTINGS_ENCRYPTION_KEY, ENTERPRISE_CLIENT_ID, ENTERPRISE_CLIENT_SECRET, ENTERPRISE_USERNAME, ENTERPRISE_PASSWORD, ENTERPRISE_PARTNER_ID, PREVIEW_SECRET, REVALIDATE_SECRET, NEXTJS_URL
Backend local	backend/.env	NODE_ENV, PORT, HOST, API_PREFIX, API_VERSION, BASE_URL, DATABASE_URL, JWT_ACCESS_SECRET, JWT_SECRET, JWT_ACCESS_EXPIRE, JWT_REFRESH_SECRET, JWT_REFRESH_EXPIRE, CORS_ORIGIN, CORS_CREDENTIALS, FRONTEND_URL, PREVIEW_SECRET, REVALIDATE_SECRET, NEXTJS_URL, RATE_LIMIT_WINDOW_MS, RATE_LIMIT_MAX_REQUESTS, RATE_LIMIT_ENABLED, FORCE_HTTPS, STORAGE_DRIVER, MAX_FILE_SIZE, UPLOAD_DIR, AWS_REGION, AWS_S3_BUCKET, AWS_S3_PUBLIC_URL, AWS_S3_ENDPOINT, AZURE_STORAGE_CONNECTION_STRING, AZURE_STORAGE_ACCOUNT_NAME, AZURE_STORAGE_CONTAINER_NAME, AZURE_STORAGE_CONTAINER_NAME, AZURE_KEY_VAULT_URL, AZURE_TENANT_ID, AZURE_CLIENT_ID, AZURE_CLIENT_SECRET,

		MFA_ENABLED, MFA_APP_NAME, LOG_LEVEL, LOG_DIR, ENABLE_FILE_LOGS, ENTERPRISE_CLIENT_ID, ENTERPRISE_CLIENT_SECRET, ENTERPRISE_USERNAME, ENTERPRISE_PASSWORD, ENTERPRISE_PARTNER_ID
Admin frontend example	frontend/.env.example	NEXT_PUBLIC_AUTH_ENABLED, NEXT_PUBLIC_API_URL, NEXT_PUBLIC_SITE_URL
Admin frontend local	frontend/.env	NEXT_PUBLIC_API_URL, NEXT_PUBLIC_AUTH_ENABLED, NEXT_PUBLIC_SITE_URL
Public web example	web/.env.example	NEXT_PUBLIC_API_URL
Public web local	web/.env	Not Found in Current Codebase
Filemanager example	filemanager/.env.example	PORT, NODE_ENV, AWS_REGION, AWS_BUCKET_NAME, CDN_URL, MAX_FILE_SIZE_MB, ALLOWED_ORIGINS, API_KEY

12. Deployment & Infrastructure

Area	Implementation Found
Deployment architecture	Separate deployable containers for backend, admin frontend, public web, and standalone filemanager.
Docker/container usage	Dockerfile found in backend, frontend, web, and filemanager; Next.js apps use standalone output.
CI/CD indication	.github/workflows/ci-build.yml builds all components; trivy-scan.yml performs vulnerability and secret scan.
Server requirement	Node.js container base images found as node:22-alpine; backend expects PostgreSQL and optional Redis/storage/cloud credentials.
Hosting/infrastructure	Backend Kubernetes manifests, ingress TLS, HPA, Key Vault CSI, securityContext, probes. Complete production values are placeholders in YAML.
Storage usage	Local uploads and cloud object storage support; S3 service and Azure Blob service present.
AWS/S3 usage	AWS SDK v3 client, bucket/region/public URL/env endpoint; upload/delete/list/presigned URL service.

13. Logging & Error Handling

Area	Implementation Found
Logging mechanism	HTTP logger middleware, request ID middleware, utils/logger, optional file logging env, activityLogger middleware/service.
Exception handling	Global errorHandler maps AppError, Prisma known errors, JWT errors, and returns standardized JSON.
Monitoring implementation	GA4 service for analytics and internal dashboard/visitor tracking. Dedicated APM/metrics exporter Not Found in Current Codebase.
Audit trail	autoLogActivity logs non-GET API operations with before/after metadata redaction; LogActivity CMS routes support list/stats/timeline/delete/cleanup.

14. Third Party Integration

Integration	Implementation Found	Configuration
Google Analytics 4	@google-analytics/data service queries GA4 reports with cache.	GA4_PROPERTY_ID, GOOGLE_SERVICE_ACCOUNT_JSON or GA4_CLIENT_EMAIL/GA4_PRIVATE_KEY.
AWS S3	S3 client/service for upload, delete, list, presigned URL.	AWS_REGION, AWS_S3_BUCKET, AWS_S3_PUBLIC_URL, AWS_S3_ENDPOINT, AWS credentials/default provider chain.
Azure Blob Storage	Azure storage service with upload/delete/public URL support.	AZURE_STORAGE_CONNECTION_STRING or AZURE_STORAGE_ACCOUNT_NAME/CONTAINER.
Azure Key Vault	Secret lookup service with DefaultAzureCredential or service principal credentials; Kubernetes CSI manifests.	AZURE_KEY_VAULT_URL, AZURE_TENANT_ID, AZURE_CLIENT_ID, AZURE_CLIENT_SECRET.
Redis/Bull	Optional Redis connection and queue-based activity/form dispatch.	REDIS_ENABLED and Redis connection envs.
Email service	Nodemailer dependency and auth/password email flows found.	SMTP/mail env details are not fully documented in inspected .env example; exact provider Not Found in Current Codebase.
Yahoo Finance	Public web Next.js API routes use yahoo-finance2 for quote and historical data.	No external credentials required by library in code.
Linknet Enterprise	Public backend proxy endpoints for coverage, nearest, cities.	ENTERPRISE_CLIENT_ID, ENTERPRISE_CLIENT_SECRET, ENTERPRISE_USERNAME, ENTERPRISE_PASSWORD, ENTERPRISE_PARTNER_ID.
Linknet Media	Public backend proxy endpoint for media data.	Exact upstream configuration Not Found in Current Codebase from route file alone.
Payment gateway	Not Found in Current Codebase.	Not Implemented.

OAuth provider	Not Found in Current Codebase.	Not Implemented.
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15. Technical Risks & Recommendations

Severity	Finding	Recommendation
High	CMS contact export route conflict	In backend/src/routes/cms/contactus.routes.ts, GET /export is declared after GET /:id, so /api/v1/cms/contactus/export may be captured as id. Move /export before /:id.
High	Frontend/backend settings endpoint mismatch	frontend settings service references /cms/settings/key/:key and /cms/settings/bulk-update, while backend defines /cms/settings/:key and /cms/settings/update-group. Align service or backend routes before SIT/UAT.
High	Frontend report grid-items endpoint mismatch	frontend report service references /cms/report-types/:id/grid-items and grid-items/update-order; backend report.routes.ts does not implement these endpoints.
Medium	JWT returned to client and localStorage usage	Backend sets HttpOnly cookies but also returns tokens in JSON, and admin frontend stores tokens in localStorage. This increases impact of XSS. Prefer cookie-only or memory-only access token strategy.
Medium	Storage abstraction inconsistency	S3 service is implemented, but storage/s3.storage.ts provider is documented as not yet implemented/throws. Complete provider or prevent STORAGE_DRIVER=s3 from selecting incomplete abstraction.
Medium	File malware scanning is opportunistic	If scanner/ClamAV is unavailable, code logs warning and allows file. For strict security review, enforce scanner availability in production.
Medium	Standalone filemanager API key optional	filemanager auth middleware disables protection if API_KEY is not configured. Production deployment must configure API_KEY or isolate the service.
Medium	Duplicate event admin aliases	Event routes expose both /cms/events and /admin/events aliases. Maintain only one canonical path or document both to avoid inconsistent frontend/API usage.
Medium	Development rate limit default	Global rate limit can be disabled outside production. Ensure RATE_LIMIT_ENABLED=true or equivalent controls in staging/prod.
Low	Backup route/controller files	Several .bak files are present in backend routes/controllers/services/validators. Exclude from build and clean up stale files before handover.
Low	Local .env files present	Local .env files exist in workspace. Ensure they are not committed and rotate any exposed credentials before deployment.
Low	Rendered ERD artifact not found	Prisma schema contains enough relation metadata, but a generated ERD diagram was Not Found in Current Codebase. Generate one for audit package if required.

16. Appendix

16.1 Important Routes

Area	Route
Backend root	GET /
Backend health	GET /health, GET /ready, GET /env-check, GET /health/detailed
API root	GET /api/v1
Auth login	POST /api/v1/auth/login
Auth refresh	POST /api/v1/auth/refresh
Admin users	/api/v1/cms/users
Admin roles	/api/v1/cms/roles
CMS pages	/api/v1/cms/pages
Public page	/api/v1/pages/:slug
Public news	/api/v1/public/news
Public forms	/api/v1/forms/:businessUnit/:slug
Upload	/api/v1/upload
File manager	/api/v1/filemanager
Standalone filemanager	/api/upload, /api/files, /api/file, /api/signed-url

16.2 Admin Frontend Page Routes

Route	Source
/bar-chart	frontend/src/app/(admin)/(others-pages)/(chart)/bar-chart/page.tsx
/line-chart	frontend/src/app/(admin)/(others-pages)/(chart)/line-chart/page.tsx
/form-elements	frontend/src/app/(admin)/(others-pages)/(forms)/form-elements/page.tsx
/basic-tables	frontend/src/app/(admin)/(others-pages)/(tables)/basic-tables/page.tsx
/blank	frontend/src/app/(admin)/(others-pages)/blank/page.tsx
/calendar	frontend/src/app/(admin)/(others-pages)/calendar/page.tsx
/profile	frontend/src/app/(admin)/(others-pages)/profile/page.tsx
/alerts	frontend/src/app/(admin)/(ui-elements)/alerts/page.tsx
/avatars	frontend/src/app/(admin)/(ui-elements)/avatars/page.tsx
/badge	frontend/src/app/(admin)/(ui-elements)/badge/page.tsx
/buttons	frontend/src/app/(admin)/(ui-elements)/buttons/page.tsx
/images	frontend/src/app/(admin)/(ui-elements)/images/page.tsx
/modals	frontend/src/app/(admin)/(ui-elements)/modals/page.tsx
/videos	frontend/src/app/(admin)/(ui-elements)/videos/page.tsx
/announcements/announcement-items	frontend/src/app/(admin)/announcements/announcement-items/page.tsx
/announcements/announcement-sections	frontend/src/app/(admin)/announcements/announcement-sections/page.tsx
/announcements/announcement-types	frontend/src/app/(admin)/announcements/announcement-types/page.tsx
/announcements	frontend/src/app/(admin)/announcements/page.tsx
/awards	frontend/src/app/(admin)/awards/page.refactored.tsx
/awards	frontend/src/app/(admin)/awards/page.tsx
/careers/:id/edit	frontend/src/app/(admin)/careers/[id]/edit/page.tsx
/careers/create	frontend/src/app/(admin)/careers/create/page.tsx
/careers	frontend/src/app/(admin)/careers/page.tsx
/cms/news/create	frontend/src/app/(admin)/cms/news/create/page.tsx
/cms/news/edit/:id	frontend/src/app/(admin)/cms/news/edit/[id]/page.tsx
/component-visibility	frontend/src/app/(admin)/component-visibility/page.tsx
/contact-data-bank	frontend/src/app/(admin)/contact-data-bank/page.tsx
/cookie-consents	frontend/src/app/(admin)/cookie-consents/page.tsx
/data/label/:parent	frontend/src/app/(admin)/data/label/[parent]/page.tsx
/data/label	frontend/src/app/(admin)/data/label/page.tsx
/events/create	frontend/src/app/(admin)/events/create/page.tsx
/events/edit/:id	frontend/src/app/(admin)/events/edit/[id]/page.tsx
/events	frontend/src/app/(admin)/events/page.tsx
/file-manager	frontend/src/app/(admin)/file-manager/page.tsx
/form-modules/:id	frontend/src/app/(admin)/form-modules/[id]/page.tsx
/form-modules/enterprise	frontend/src/app/(admin)/form-modules/enterprise/page.tsx
/form-modules/fiber	frontend/src/app/(admin)/form-modules/fiber/page.tsx
/form-modules/media	frontend/src/app/(admin)/form-modules/media/page.tsx
/form-modules	frontend/src/app/(admin)/form-modules/page.tsx
/form-modules/submissions	frontend/src/app/(admin)/form-modules/submissions/page.tsx
/form-submissions/:formModuleId/:submissionId	frontend/src/app/(admin)/form-submissions/[formModuleId]/[submissionId]/page.tsx
/form-submissions	frontend/src/app/(admin)/form-submissions/page.tsx
/log-activity	frontend/src/app/(admin)/log-activity/page.tsx
/management/:categoryId	frontend/src/app/(admin)/management/[categoryId]/page.tsx
/management	frontend/src/app/(admin)/management/page.tsx
/menu-management	frontend/src/app/(admin)/menu-management/page.tsx
/mfa-setup	frontend/src/app/(admin)/mfa-setup/page.tsx
/news/category	frontend/src/app/(admin)/news/category/page.tsx
/news/data	frontend/src/app/(admin)/news/data/page.tsx
/news/highlight	frontend/src/app/(admin)/news/highlight/page.tsx
/	frontend/src/app/(admin)/page.tsx
/pages/:id	frontend/src/app/(admin)/pages/[id]/page.tsx
/pages/create	frontend/src/app/(admin)/pages/create/page.tsx
/pages	frontend/src/app/(admin)/pages/page.tsx
/reports	frontend/src/app/(admin)/reports/page.tsx
/reports/report-items	frontend/src/app/(admin)/reports/report-items/page.tsx
/reports/report-sections	frontend/src/app/(admin)/reports/report-sections/page.tsx
/reports/report-types	frontend/src/app/(admin)/reports/report-types/page.tsx
/roles-permissions	frontend/src/app/(admin)/roles-permissions/page.tsx
/settings	frontend/src/app/(admin)/settings/page.tsx
/url-redirection	frontend/src/app/(admin)/url-redirection/page.tsx

/users-management	frontend/src/app/(admin)/users-management/page.tsx
/signin	frontend/src/app/(full-width-pages)/(auth)/signin/page.tsx
/signup	frontend/src/app/(full-width-pages)/(auth)/signup/page.tsx
/error-404	frontend/src/app/(full-width-pages)/(error-pages)/error-404/page.tsx
/login	frontend/src/app/(full-width-pages)/login/page.tsx
/mfa-verify	frontend/src/app/(full-width-pages)/mfa-verify/page.tsx
/linknet-enterprise-coverage	frontend/src/app/linknet-enterprise-coverage/page.tsx
/linknet-media	frontend/src/app/linknet-media/page.tsx
/page/:slug	frontend/src/app/page/[slug]/page.tsx

16.3 Public Web Page Routes

Route	Source
/:locale/:slug	web/app/[locale]/[...slug]/page.tsx
/:locale/career/:id	web/app/[locale]/career/[id]/page.jsx
/:locale/career	web/app/[locale]/career/page.jsx
/:locale/enterprise/form/success	web/app/[locale]/enterprise/form/success/page.jsx
/:locale/events/:slug	web/app/[locale]/events/[slug]/page.jsx
/:locale/events	web/app/[locale]/events/page.jsx
/:locale/life-at-linknet	web/app/[locale]/life-at-linknet/page.jsx
/:locale/news/:slug	web/app/[locale]/news/[slug]/page.jsx
/:locale/news/category/:slug	web/app/[locale]/news/category/[slug]/page.jsx
/:locale/news	web/app/[locale]/news/page.jsx
/:locale	web/app/[locale]/page.js
/:locale/section	web/app/[locale]/section/page.jsx

16.4 Important Commands

Component	Command	Purpose
Backend	npm install	Install dependencies.
Backend	npx prisma generate	Generate Prisma client.
Backend	npx prisma migrate deploy	Apply migrations in deployed environment.
Backend	npm run build	Compile TypeScript.
Backend	npm start	Run compiled backend server.
Admin Frontend	npm run dev	Run admin app in development.
Admin Frontend	npm run build	Build admin Next.js app.
Public Web	npm run dev	Run public web in development.
Public Web	npm run build	Build public Next.js app.
Filemanager	npm run build	Compile filemanager service.
Filemanager	npm start	Run filemanager service.
Docker	docker build -t <image> <component-dir>	Build component image using found Dockerfile.